

Rate, Leakage, and Distortion for Gaussian Sources: Water-Filling on a Tilted Weight

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Abstract

An encoder observes a jointly Gaussian vector and describes part of it to a decoder under a weighted quadratic distortion. A second party holds a noisy linear observation of the source and never decodes. The description then has two costs, its rate and its conditional leakage to the second party, which is also the least rate at which that party can re-encode it. The single-letter region is known from secure source coding, and we determine its structure. In every geometry and at every budget, each frontier point is the fixed point of reverse water-filling on the decoder's weight tilted by the second party's information, and for a scalar second-party observation it reduces to two scalar equations. For a scalar described variable the frontier is a one-parameter curve of projection directions, located by one scalar root. The direction of the best one-dimensional description turns with the budget unless the two parties' geometries share it, while a description built for a single party keeps fixed directions, with an explicit lower bound on its excess leakage. When the optimal description is one-dimensional, which an explicit test certifies, the minimum leakage is a closed-form function of the largest eigenvalue of a matrix pencil in which the budget enters. The tradeoff exists because the distortion is a trace-type constraint. Under a matrix distortion on the whole source, one description minimizes both costs. When the decoder also has side information, we prove the region for the continuous source and settle the two degraded cases. If the decoder knows more than the second party there is no tradeoff, and if it knows less the problem reduces exactly to the one without decoder side information.

Index Terms

Equivocation, Gaussian sources, information leakage, rate-distortion theory, secure source coding, side information.

I. INTRODUCTION

One description of data is often held by more than one party. The decoder that reconstructs from it pays its rate. A second party that holds the description together with its own observation of the source learns from it the conditional mutual information between the source and the description given that observation. In a secrecy reading the second party is an eavesdropper and this quantity is the information leakage beyond its side information. Minimizing it maximizes the eavesdropper's equivocation about the source. In a storage reading the second party keeps the description compressed against what it already knows, and the same quantity is the least rate at which it can re-encode the description. Either way one description carries two costs, set by two different parties, and the encoder, which may observe more of the source than either party, decides how they trade off.

The single-letter region of jointly achievable rate, distortion, and equivocation is due to Villard and Piantanida [1, Theorem 3], and for vector Gaussian sources Ekrem and Ulukus [2] obtained the minimum leakage when the rate is unconstrained, as an optimization over covariance matrices. The joint rate-leakage region was left open. They report that they could not solve the weighted problem that traces it [2, eqs. (30)–(31)], and Xu, Chen, Chen, and Jin [5] have since proved by a new extremal inequality that Gaussian auxiliaries solve it, leaving the region as a matrix program. Without decoder side information the region itself is elementary, because a single Gaussian exhaustion lemma bounds both costs at once (Proposition 2). What this paper contributes is the structure of that region for an informed encoder. The encoder observes a jointly Gaussian vector T , the decoder must reproduce a linear part $Y = FT$ under a weighted quadratic distortion, and the second party observes $S = HT + U$ for an arbitrary matrix H . The two costs disagree because the distortion is a single trace-type constraint, which leaves the encoder a choice among error covariances. Under a matrix distortion on the whole source, the formulation of the earlier work, one description minimizes both costs, as the bounds of Ekrem and Ulukus already imply (Proposition 4). Which

directions the optimal description favors is set by how the decoder's weight and the second party's information are aligned, whether or not the encoder observes more than it describes. When H also reaches parts of T outside Y , the encoder holds information about the second party's observation that the decoder does not need, and then the leakage is not determined by the joint law of the described variable and the second party's observation.

The results are structural as well as explicit. The coding theorem (Theorem 3) identifies the operational region with the single-letter one, and Gaussian descriptions, indexed by their error covariance, exhaust it (Proposition 2). In that coordinate both costs are convex (Theorem 5), so every point of the rate–leakage frontier is the unique minimizer of a convex program and the minimum leakage is a determinant maximization under linear matrix inequalities. When the optimal description is one-dimensional, which an explicit certificate decides, the minimum leakage is

$$L(D) = \frac{1}{2} \log(1 + \Delta/\mu_{\max}),$$

with Δ the reduction in distortion the description must buy and $\mu_{\max}$ the largest generalized eigenvalue of a pencil formed from the source covariance, the decoder's weight, the conditional covariance left after the second party's observation, and the budget (Theorem 16). For a scalar described variable and a scalar context this is an explicit closed form (Corollary 17). When the decoder's and the holder's geometries commute, the whole frontier is explicit at every budget (Theorem 13), and more generally the problem splits along every subspace decomposition that both geometries respect (Theorem 12). When the decoder reproduces the whole source, the frontier is explicit at every small budget in any geometry, in the eigenbasis of the second party's information relative to the decoder's weight (Theorem 15). In general an optimal description has no more components than the described variable, even though it uses the context (Proposition 6). The decoupling, commuting, and low-distortion results are cases of one characterization that holds at every budget, in every geometry, for partial and full descriptions alike. In it the error covariance is the fixed point of reverse water-filling on the spectrum of the decoder's weight tilted by the information the second party's observation adds to the description (Theorem 7). For a scalar observation the tilt has rank one, and the whole frontier reduces to two scalar equations (Corollary 9). Every frontier test channel is also the rate–distortion optimal test channel for a decoder that must describe the holder's view as well as Y , under a weight fixed by the solution (Corollary 8). For a scalar described variable, with any context and any holder, the frontier is explicit up to one scalar root. Every optimal description reads a direction on the curve $((1 - \varrho)\Sigma_T + \varrho\Sigma_{T|S})^{-1}\Sigma_T q$, $\varrho \in [0, 1]$, and one scalar equation, whose root is unique for the minimum leakage, selects it (Theorem 18). For holders and described variables of any rank, the frontier is the limit of an iteration of classical reverse water-filling steps that decreases the cost monotonically (Proposition 11).

The eigenvector of the pencil is the direction the description reads, and it carries the paper's main structural finding. The rate-optimal description reads fixed directions at every budget, the principal direction of the decoder's weighted covariance while it is one-dimensional, as do principal components, reverse water-filling [3], and the Gaussian information bottleneck [4]. The best one-dimensional description, which is the leakage-optimal one wherever the certificate of Theorem 16(b) holds, reads a direction that turns with the budget, unless that direction is a common generalized eigenvector of the decoder's geometry and the second party's (Theorem 20). No description optimized for a single party under a weighted quadratic distortion does this (Proposition 19), and a one-dimensional description with a fixed projection direction pays an explicit excess leakage over the best one-dimensional description, at one of any two budgets whose optimal directions differ (Theorem 25). The same condition governs whether the two optimal descriptions coincide at all (Theorem 22). While the rate-optimal description is one-dimensional, they differ whenever the second party's information does not map the decoder's principal direction onto itself. For independent components the leakage-optimal split of the distortion equalizes slopes that we compute in closed form, activates components in an order that is not the order of their variances, and can describe a disjoint set of components from the rate-optimal split (Section VIII). With side information at the decoder (Section IX) we prove the single-letter region for the continuous source and solve the two degraded cases. When the decoder's information dominates the holder's, the leakage equals the rate at the optimum and there is no tradeoff. When the holder's dominates, an estimation inequality (Lemma 30) reduces the problem exactly to the one without decoder side information, for the source conditioned on the decoder's observation and the holder's excess information. The same effects appear in fixed-rate scalar quantizers (Appendix C). At equal rate and under the same distortion budget, which the principal-axis quantizer meets with slack, a Lloyd–Max quantizer applied to a leakage-aware projection leaks 13% to 72% less than the same quantizer on the principal axis, and its projection direction turns with the budget.

Related work

The conditional leakage studied here is the secrecy literature's quantity. For a deterministic encoder, $h(T^n | M, S^n) = h(T^n | S^n) - H(M | S^n)$, so minimizing $H(M | S^n)/n$, which equals $I(T^n; M | S^n)/n$, is the same as maximizing the equivocation $h(T^n | M, S^n)/n$. Secure lossy source coding with side information at the eavesdropper was characterized by Villard and Piantanida [1]. With no decoder side information their Theorem 3 gives the discrete region of Theorem 32 below. Ekrem and Ulukus [2] treat vector Gaussian sources with side information at both receivers, with arbitrary observation maps [2, Sec. 5], under a matrix distortion on the whole source. They give an outer bound that is tight without a rate constraint and leave the joint region open [2, eqs. (30)–(31)]. Their leakage is the total information at the eavesdropper, which differs from the conditional leakage used here by the constant $I(T; S)$. Without decoder side information, the region of Proposition 2 and Theorem 3 is the union of their matrix-distortion regions over $\Lambda \preceq \Sigma_T$ with $\text{tr}(Q\Lambda) \leq D$, because a code meeting the trace constraint may output $\mathbb{E}[T | M]$, whose error matrix is at most Σ_T . Appendix A proves the continuous-alphabet coding theorem directly. Xu, Chen, Chen, and Jin [5] prove, by a new extremal inequality, that Gaussian auxiliaries are optimal in the Ekrem–Ulukus problem for every weight, which characterizes the rate–distortion–equivocation function of vector Gaussian secure source coding. In the preprint version [6], which we checked, the characterization is a matrix program over two precision matrices [6, eq. (189), Thm. 4] for full-rank side informations at both receivers. Absent decoder side information, a formal limit outside their theorem, the program reduces to the one of Proposition 2 in the precision coordinate $\Sigma_{e0}^{-1} - \Sigma_T^{-1}$ with the matrix constraint, which has no tradeoff (Proposition 4). These results fix the form of the optimizers. The structure of the solution under a trace-type distortion, which is the subject here, is not addressed there. Günlü, Schaefer, Boche, and Poor [11] evaluate the lossy rate region for scalar Gaussian sources with a remote source and separate secrecy and privacy leakage. None of these works treats an encoder that observes components it need not describe, a trace-type distortion on part of the source, or the structure results below. Related problems price the leakage of a description about side information [12], [13], [14] or the equivocation of the source in a cipher system [15], [16]. The leakage endpoint is Steinberg's common-reconstruction functional [17] on the source as the encoder observes it, an augmentation noted for finite alphabets by Lapidath, Malär, and Wigger [18]. A vector Gaussian common-reconstruction function with every component penalized was given by Lu and Xu [19]. Weighted quadratic distortion models a decoder that cares about some directions more than others, as in weighted rate–distortion [20], indirect source coding [21], and task-based quantization [22]. Parameterizing a Gaussian description by its error covariance is a standard device in vector Gaussian source coding [23], and the resulting endpoint program is a determinant maximization [24].

The paper is organized as follows. Section II sets up the model and the coding theorem, Section III proves the convex structure, Section IV gives the water-filling characterization of the whole frontier, Section V its explicit cases, Section VI the one-dimensional case, Section VII the results on directions and misalignment, Section VIII independent components, and Section IX decoder side information. Section X gives instances and Section XI open problems. The appendices prove the coding theorem and a Gaussian exhaustion lemma, evaluate practical scalar quantizers (Appendix C), and describe the numerical checks.

II. MODEL AND THE CODING THEOREM

Logarithms are base two unless written $\ln$. The encoder observes i.i.d. copies of $T \in \mathbb{R}^p$, jointly Gaussian and centered with $\Sigma_T \succ 0$. The described variable is $Y = FT \in \mathbb{R}^m$ for a matrix F of full row rank, and the decoder's distortion is $\mathbb{E}[(Y - \hat{Y})^\top W(Y - \hat{Y})]$ for a weight $W \succ 0$. Write

$$Q := F^\top W F \succeq 0, \quad D_{\max} := \text{tr}(Q\Sigma_T).$$

The second party, which we call the holder, observes $S = HT + U \in \mathbb{R}^r$ with $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, independent of T and of all coding randomness. Write

$$J := H^\top \Sigma_U^{-1} H \succeq 0, \quad \Sigma_{T|S} = (\Sigma_T^{-1} + J)^{-1},$$

the information that S carries about T and the conditional covariance it leaves.

A code consists of an encoder $f_n : \mathbb{R}^{p \times n} \rightarrow \mathcal{M}_n$, a finite set, and a decoder $g_n : \mathcal{M}_n \rightarrow \mathbb{R}^{m \times n}$. With $M = f_n(T^n)$ and $\hat{Y}^n = g_n(M)$, its rate, distortion, and leakage are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(Y_i - \hat{Y}_i)^\top W(Y_i - \hat{Y}_i)], \quad L_n = \frac{1}{n} H(M | S^n) = \frac{1}{n} I(T^n; M | S^n).$$

A pair (R, L) is achievable at distortion D when a sequence of codes has $\limsup R_n \leq R$, $\limsup L_n \leq L$, and $\limsup D_n \leq D$. The closure of the achievable set is $\mathcal{R}(D)$. Throughout $0 < D < D_{\max}$. At $D \geq D_{\max}$ the zero reproduction gives $R = L = 0$. Write $R_{\min}(D)$ and $L(D)$ for the least rate and the least leakage over $\mathcal{R}(D)$, and call a description rate-optimal or leakage-optimal when it attains one of them.

A test channel $p(\hat{y} | t)$ is admissible when its distortion is at most D . The Markov chain $\hat{Y} - T - S$ holds because the description is formed without S . For a channel, let Σ_{e0} and Σ_e be the error covariances of the best linear estimates of T from $\hat{Y}$ and from $(\hat{Y}, S)$. Appendix B proves the following Gaussian exhaustion lemma.

Lemma 1. *Let $\hat{Y} \in \mathbb{R}^k$, for any k , be centered with finite second moments, let A be its regression matrix on T and $N' = \hat{Y} - AT$ its residual, and suppose $\text{Cov}(N', S) = 0$ and $\text{Cov } N' \succ 0$. Then $I(T; \hat{Y}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$ and $I(T; \hat{Y} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the jointly Gaussian channel $AT + N$, $N \sim \mathcal{N}(0, \text{Cov } N')$ independent of (T, U) , has the same second moments and meets both bounds with equality.*

Proposition 2. *The union of the quadrants above $(I(T; \hat{Y}), I(T; \hat{Y} | S))$ over admissible channels equals the union, over $\Sigma_{e0} \in \mathcal{P}(D)$, of the quadrants above*

$$(r(\Sigma_{e0}), \ell(\Sigma_{e0})) = \left(\frac{1}{2} \log \frac{\det \Sigma_T}{\det \Sigma_{e0}}, \frac{1}{2} \log \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \right), \quad \Sigma_e^{-1} = \Sigma_{e0}^{-1} + J, \quad (1)$$

where $\mathcal{P}(D) = \{\Sigma_{e0} : 0 \prec \Sigma_{e0} \preceq \Sigma_T, \text{tr}(Q\Sigma_{e0}) \leq D\}$.

Proof. Given an admissible channel, centering $\hat{Y}$ changes neither coordinate and does not raise the distortion. If $\text{Cov } N'$ is singular, either a noiseless linear functional of T is revealed and both coordinates are infinite, or a nonzero combination of the components of $\hat{Y}$ vanishes and may be dropped after an invertible linear map. So assume $\text{Cov } N' \succ 0$. The residual is uncorrelated with T by construction and with U by independence, hence with S , and Lemma 1 replaces the channel by $\hat{Y} = AT + N$ with the same distortion and no larger coordinates. For it, $\Sigma_{e0}^{-1} = \Sigma_T^{-1} + A^\top \Sigma_N^{-1} A$, so $0 \prec \Sigma_{e0} \preceq \Sigma_T$. The pair $(\hat{Y}, S)$ is a linear observation of T with independent noises, so $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J$. The distortion is at least that of $\mathbb{E}[Y | \hat{Y}]$, which is $\text{tr}(Q\Sigma_{e0})$. Conversely, given $\Sigma_{e0} \in \mathcal{P}(D)$, factor $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = G^\top G$, let $Z = GT + N$ with $N \sim \mathcal{N}(0, I)$ independent of (T, U) , and reproduce $\hat{Y} = \mathbb{E}[Y | Z]$. Then $\text{Cov}(T | Z) = \Sigma_{e0}$, the distortion is $\text{tr}(Q\Sigma_{e0}) \leq D$, $\hat{Y} - T - S$ holds, and since $\hat{Y}$ is a function of Z , data processing bounds its coordinates by those of Z , which are (1). $\square$

Theorem 3. *For every $0 < D < D_{\max}$, $\mathcal{R}(D)$ equals the union of Proposition 2.*

The proof is in Appendix A. Its discrete core is the region of Villard and Piantanida without decoder side information (Theorem 32). The Gaussian extension quantizes the side information, the source, and the reproduction and passes to the limit.

The tradeoff studied below needs the distortion to leave the encoder a choice of error covariance. The vector Gaussian secrecy literature [2], [5] instead asks the decoder to reproduce the whole source under a matrix constraint, and without decoder side information that choice disappears. The bounds of Ekrem and Ulukus imply the following. We state it in the present notation because the contrast organizes the rest of the paper.

Proposition 4 (cf. [2, Thms. 6–7]). *Suppose the decoder reproduces all of T , an admissible channel is one whose reproduction $\hat{T}$ has $\mathbb{E}[(T - \hat{T})(T - \hat{T})^\top] \preceq \Lambda$, and $0 \prec \Lambda \preceq \Sigma_T$. Then the union of the quadrants above $(I(T; \hat{T}), I(T; \hat{T} | S))$ over admissible channels is the single quadrant above $(r(\Lambda), \ell(\Lambda))$ of (1), so one description is both rate-optimal and leakage-optimal.*

Proof. The proof of Proposition 2 applies with the matrix constraint in place of the trace, because the error matrix of $\hat{T}$ depends only on second moments and dominates $\text{Cov}(T | \hat{T}) = \Sigma_{e0}$ for the Gaussian channel. The feasible set becomes $\{0 \prec \Sigma_{e0} \preceq \Lambda\}$. Both r and ℓ are nonincreasing in the Loewner order, since $\Sigma_e = (\Sigma_{e0}^{-1} + J)^{-1}$ is monotone in Σ_{e0} , and Λ is the largest feasible element. $\square$

Under the weighted constraint $\text{tr}(Q\Sigma_{e0}) \leq D$ the feasible set has no largest element, and which maximal element the encoder picks is the whole problem. The same is true of a matrix constraint on the described part Y alone, $F\Sigma_{e0}F^\top \preceq \Lambda_Y$, which leaves the error covariance free outside Y .

Running example. Two instances recur. The scalar instance has $T = (Y, V)$ with unit variances and correlation ρ , $W = 1$, and $S = V + U$ with $\text{Var } U = \tau^2$. The coupled instance has a two-dimensional described variable with

$\Sigma_Y = \text{diag}(1, 0.4)$, $W = I$, a scalar context with $\text{Var } V = 1$ and $\text{Cov}(Y, V) = 0.6u$ for the unit vector u at 50° , and $S = V + U$ with $\text{Var } U = 0.05$, so $D_{\max} = 1.4$. In both the holder observes only the context.

III. CONVEX STRUCTURE

Theorem 5. *On $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$, the rate r and the leakage ℓ of (1) are strictly convex functions of Σ_{e0} , and the distortion $\text{tr}(Q\Sigma_{e0})$ is linear. Consequently the following hold.*

- (a) *For every $\alpha \in [0, 1]$ the program $\min\{\alpha r(\Sigma_{e0}) + (1 - \alpha)\ell(\Sigma_{e0}) : \Sigma_{e0} \in \mathcal{P}(D)\}$ has a unique minimizer, the distortion constraint is active there, and the minima over α trace the Pareto frontier of $\mathcal{R}(D)$.*
- (b) *In the coordinate Σ_e the minimum leakage is a determinant maximization,*

$$L(D) = \frac{1}{2} \log \det \Sigma_{T|S} - \frac{1}{2} \max\{\log \det \Sigma_e : 0 \prec \Sigma_e \preceq \Sigma_{T|S}, \text{tr}(Qh(\Sigma_e)) \leq D\}, \quad h(\Sigma_e) := (\Sigma_e^{-1} - J)^{-1}, \quad (2)$$

in which h is matrix-convex and the constraint is equivalent to a linear matrix inequality. With $J = B^\top B$, $\text{tr}(Qh(\Sigma_e)) \leq D$ if and only if some symmetric Z has $\text{tr}(QZ) \leq D$ and

$$\begin{bmatrix} Z - \Sigma_e & \Sigma_e B^\top \\ B \Sigma_e & I - B \Sigma_e B^\top \end{bmatrix} \succeq 0; \quad (3)$$

- (c) *A feasible Σ_e is the maximizer in (2) if and only if $\text{tr}(Qh(\Sigma_e)) = D$ and there are $\varpi \geq 0$ and $\Theta \succeq 0$ with*

$$\frac{1}{2} \Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0} = \Sigma_e \Theta \Sigma_e, \quad \Theta(\Sigma_{T|S} - \Sigma_e) = 0, \quad \Sigma_{e0} = h(\Sigma_e), \quad (4)$$

and then $\varpi > 0$.

Proof. Convexity. $r = \frac{1}{2} \log \det \Sigma_T - \frac{1}{2} \log \det \Sigma_{e0}$ is strictly convex. For the leakage, write $J = B^\top B$ and note that $\Sigma_e = (\Sigma_{e0}^{-1} + B^\top B)^{-1}$ is the error covariance of the best linear estimate of a vector with covariance Σ_{e0} from its noisy image under B with unit noise. For every matrix L , the estimator with gain L has error covariance $E_L(\Sigma_{e0}) = (I - LB)\Sigma_{e0}(I - LB)^\top + LL^\top \succeq \Sigma_e$, with equality at the optimal gain, and E_L is affine in Σ_{e0} . A pointwise minimum of affine maps in the Löwner order, attained at each point, is matrix-concave. At $\Sigma_t = (1 - t)\Sigma_0 + t\Sigma_1$ with optimal gain L_t , $\Sigma_e(\Sigma_t) = E_{L_t}(\Sigma_t) = (1 - t)E_{L_t}(\Sigma_0) + tE_{L_t}(\Sigma_1) \succeq (1 - t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)$. The map $\Sigma \mapsto -\log \det \Sigma$ is strictly convex and decreasing in the Löwner order, so $-\log \det \Sigma_e(\Sigma_t) \leq -\log \det((1 - t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)) \leq (1 - t)(-\log \det \Sigma_e(\Sigma_0)) + t(-\log \det \Sigma_e(\Sigma_1))$, with strict inequality in the second step when $\Sigma_0 \neq \Sigma_1$, because $\Sigma_{e0} \mapsto \Sigma_e$ is injective. So ℓ is strictly convex.

(a) $\mathcal{P}(D)$ is convex, the objective is strictly convex and diverges as $\det \Sigma_{e0} \rightarrow 0$, and the feasible set is bounded, so the minimizer exists and is unique. Both r and ℓ decrease when Σ_{e0} increases in the Löwner order, so at a minimizer with slack distortion a move toward Σ_T would stay feasible and lower the objective, since Σ_T itself has distortion $D_{\max} > D$. The region is convex and upward closed, so its Pareto points are exactly the minima of the weighted programs.

(b) The map $\Sigma_{e0} \mapsto \Sigma_e$ is a bijection of $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$ onto $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$, with inverse h , since $\Sigma_{e0} \preceq \Sigma_T$ if and only if $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J \succeq \Sigma_T^{-1} + J = \Sigma_{T|S}^{-1}$. The matrix $B\Sigma_{T|S}B^\top$ has the same nonzero eigenvalues as $\Sigma_{T|S}^{1/2}J\Sigma_{T|S}^{1/2} \prec \Sigma_{T|S}^{1/2}(\Sigma_T^{-1} + J)\Sigma_{T|S}^{1/2} = I$, so on the feasible set $B\Sigma_eB^\top \prec I$, and the Woodbury identity gives $h(\Sigma_e) = \Sigma_e + \Sigma_e B^\top (I - B\Sigma_eB^\top)^{-1} B \Sigma_e$. By the Schur complement criterion [25, Sec. A.5.5], $h(\Sigma_e) \preceq Z$ if and only if (3) holds, and the block matrix there is affine in (Σ_e, Z) . Averaging two instances with $Z_i = h(\Sigma_e^{(i)})$ gives matrix convexity of h . Since $\text{tr}(QZ)$ is monotone in Z , the constraint $\text{tr}(Qh(\Sigma_e)) \leq D$ is equivalent to the existence of Z with $\text{tr}(QZ) \leq D$ and (3).

(c) The program (2) has a strictly feasible point ($\Sigma_{e0} = \epsilon \Sigma_T$ with $\epsilon D_{\max} < D$), so the Karush–Kuhn–Tucker conditions are necessary and sufficient. The differential of h is $dh = h\Sigma_e^{-1}d\Sigma_e\Sigma_e^{-1}h$, so the gradient of $\text{tr}(Qh)$ is $\Sigma_e^{-1}\Sigma_{e0}Q\Sigma_{e0}\Sigma_e^{-1}$, and stationarity of $-\frac{1}{2} \ln \det \Sigma_e + \varpi \text{tr}(Qh) + \text{tr}(\Theta(\Sigma_e - \Sigma_{T|S}))$, multiplied on both sides by Σ_e , is the first condition of (4), the factor $\ln 2$ being absorbed in ϖ . When the distortion constraint is active its complementary slackness holds for every $\varpi \geq 0$, and the maximizer has an active constraint by (a). If $\varpi = 0$, then $\Theta = \frac{1}{2}\Sigma_e^{-1} \succ 0$ and the second condition forces $\Sigma_e = \Sigma_{T|S}$, which is infeasible. So $\varpi > 0$. $\square$

The optimal description uses the context, but it never needs more dimensions than the described variable.

Proposition 6. For every $\alpha \in [0, 1]$ the minimizer Σ_{e0} of Theorem 5(a) has $\text{rank}(\Sigma_T - \Sigma_{e0}) \leq \text{rank } Q$, so it is attained by a jointly Gaussian description $GT + N$ with at most $\text{rank } Q$ components.

Proof. The weighted program in Σ_{e0} is convex with the strictly feasible point $\epsilon \Sigma_T$, so there are $\nu \geq 0$ and $\Theta \succeq 0$ with $\Theta(\Sigma_T - \Sigma_{e0}) = 0$ and, from the gradients $\nabla r = -\frac{1}{2}\Sigma_{e0}^{-1}$ and $\nabla \ell = -\frac{1}{2}\Sigma_{e0}^{-1}\Sigma_e\Sigma_{e0}^{-1}$ (in natural units),

$$\frac{1}{2}\Sigma_{e0}^{-1}(\alpha\Sigma_{e0} + (1 - \alpha)\Sigma_e)\Sigma_{e0}^{-1} = \nu Q + \Theta.$$

The left side is positive definite, so $\text{rank } \Theta \geq p - \text{rank } Q$, and the range of $\Sigma_T - \Sigma_{e0}$ lies in the null space of Θ . The precision $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = \Sigma_{e0}^{-1}(\Sigma_T - \Sigma_{e0})\Sigma_T^{-1}$ has the same rank, and factoring it as $G^\top G$ gives the description of Proposition 2. $\square$

IV. WATER-FILLING ON A TILTED WEIGHT

The frontier has one characterization that holds in every geometry and at every budget, for partial and full descriptions alike. It rests on the identity $\ell = r - I(\hat{\mathbf{Y}}; S)$, which holds because $\hat{\mathbf{Y}} - T - S$. Whiten by the source covariance,

$$X = \Sigma_T^{-1/2}\Sigma_{e0}\Sigma_T^{-1/2}, \quad \tilde{Q} = \Sigma_T^{1/2}Q\Sigma_T^{1/2}, \quad \tilde{H} = H\Sigma_T^{1/2},$$

so that $r = -\frac{1}{2}\log \det X$, $I(\hat{\mathbf{Y}}; S) = \frac{1}{2}\log \det \Sigma_S - \frac{1}{2}\log \det(\tilde{H}X\tilde{H}^\top + \Sigma_U)$, the distortion is $\text{tr}(\tilde{Q}X)$, and the cap is $X \preceq I$. For a symmetric $A \succeq 0$ with eigenpairs (λ_i, e_i) , let

$$\Phi(A) = \sum_i \min\{1, 1/\lambda_i\} e_i e_i^\top,$$

with $\min\{1, 1/\lambda_i\} = 1$ when $\lambda_i = 0$, reverse water-filling at level one on the spectrum of A .

Theorem 7. For $\alpha \in [0, 1]$ and $0 < D < D_{\max}$, the frontier point of weight α is the unique X with $\text{tr}(\tilde{Q}X) = D$ and

$$X = \Phi(\tilde{A}(X)), \quad \tilde{A}(X) = 2\nu\tilde{Q} + (1 - \alpha)\tilde{H}^\top(\tilde{H}X\tilde{H}^\top + \Sigma_U)^{-1}\tilde{H} \quad (5)$$

for some $\nu > 0$.

Proof. In natural units the weighted objective is $-\frac{1}{2}\ln \det X + \frac{1-\alpha}{2}\ln \det(\tilde{H}X\tilde{H}^\top + \Sigma_U)$ plus a constant, convex in X by Theorem 5, with gradient $-\frac{1}{2}X^{-1} + \frac{1-\alpha}{2}\tilde{H}^\top(\tilde{H}X\tilde{H}^\top + \Sigma_U)^{-1}\tilde{H}$. The program has the strictly feasible point ϵI , so a point is the minimizer if and only if $\text{tr}(\tilde{Q}X) \leq D$ and there are $\nu \geq 0$ and $\Theta \succeq 0$ with $\Theta(I - X) = 0$, $\nu(\text{tr}(\tilde{Q}X) - D) = 0$, and $X^{-1} = \tilde{A}(X) + 2\Theta$. Suppose these hold. Let $\mathcal{N}$ be the null space of $I - X$. The range of Θ lies in $\mathcal{N}$, and for $v \in \mathcal{N}$, $\tilde{A}v = v - 2\Theta v \in \mathcal{N}$, so $\mathcal{N}$ and $\mathcal{N}^\perp$ are invariant under $\tilde{A}$. On $\mathcal{N}$, $\tilde{A} = I - 2\Theta \preceq I$ and $X = I$. On $\mathcal{N}^\perp$, $\Theta = 0$, $X^{-1} = \tilde{A}$, and $X \preceq I$ gives $\tilde{A} \succeq I$. So $X = \Phi(\tilde{A})$. Conversely, if $X = \Phi(\tilde{A})$, the matrix $\Theta = \frac{1}{2}\sum_{\lambda_i < 1}(1 - \lambda_i)e_i e_i^\top$ is positive semidefinite, vanishes off the eigenvectors on which $X = I$, and satisfies $X^{-1} = \tilde{A} + 2\Theta$. If $\nu = 0$, these are the optimality conditions of the program without the distortion constraint, whose unique minimizer is $X = I$ because both costs decrease in the Löwner order. Its distortion is $D_{\max} > D$, so $\nu > 0$. The minimizer is unique by strict convexity. $\square$

The second term of $\tilde{A}$ is the information about T that the holder's observation adds to the description, weighted by $1 - \alpha$. It equals $(1 - \alpha)X^{-1}(X - X_e)X^{-1}$ with $X_e = \Sigma_T^{-1/2}\Sigma_e\Sigma_T^{-1/2}$. At $\alpha = 1$ it vanishes, and the theorem is reverse water-filling on the spectrum of $\tilde{Q}$, the weighted rate-distortion solution. For $\alpha < 1$ the holder tilts the weight, and the error covariance is reverse water-filling on the tilted spectrum. The decoupling, commuting, and low-distortion results of Section V are the cases in which $\tilde{Q}$ and the tilt leave common subspaces invariant, in which they are diagonal in a common basis, and in which $Q \succ 0$ and no eigenvalue of $\tilde{A}$ is below one. In the last case $\Phi(\tilde{A}) = \tilde{A}^{-1}$, so $X^{-1} = \tilde{A}(X)$, which by the Woodbury identity is the stationarity condition in the proof of Theorem 15.

Corollary 8. Let Σ_{e0} be the frontier point of weight $\alpha < 1$ and ν its multiplier in (5). Then Σ_{e0} is the rate-distortion optimal error covariance, at its own distortion, for the weighted quadratic distortion with weight

$$Q' = Q + \kappa H^\top (H\Sigma_{e0}H^\top + \Sigma_U)^{-1}H, \quad \kappa = \frac{1 - \alpha}{2\nu},$$

that is, for a decoder that reproduces Y under the weight W and the holder's noiseless view HT under the weight $\kappa \text{Cov}(S | \hat{\mathbf{Y}})^{-1}$, with $\text{Cov}(S | \hat{\mathbf{Y}}) = H\Sigma_{e0}H^\top + \Sigma_U$ evaluated at the frontier description itself.

Proof. In whitened coordinates the weighted rate–distortion solution for Q' at a given distortion is $\Phi(2\nu'\tilde{Q}')$ for the level ν' that meets it, which is unique. With $\tilde{Q}' = \tilde{A}(X)/(2\nu)$ and the distortion $\text{tr}(\tilde{Q}'X)$, the level $\nu' = \nu$ gives $\Phi(\tilde{A}(X)) = X$ by Theorem 7. $\square$

So the least-leaking description at a given rate is the one a rate–distortion encoder would produce if it also had to describe what the holder sees. Making the description informative about S is what lowers $\ell = r - I(\hat{\mathbf{Y}}; S)$. The weight Q' depends on the frontier point, so the corollary is an equivalence of single-letter solutions, not a construction. It says that the optimal test channel is a weighted rate–distortion test channel, and the operational consequences for codes designed for Q' are not developed here. The target (Y, HT) need not have full row rank, and nothing in the corollary uses it.

Corollary 9. Suppose the holder's observation is scalar, $S = \zeta^\top T + U$ with $\text{Var } U = \sigma_U^2$, and let $\tilde{\zeta} = \Sigma_T^{1/2}\zeta$. Then the frontier point of weight α is $X = \Phi(2\nu\tilde{Q} + \omega\tilde{\zeta}\tilde{\zeta}^\top)$, where the two scalars $\nu > 0$ and $\omega \in [(1-\alpha)/(\|\tilde{\zeta}\|^2 + \sigma_U^2), (1-\alpha)/\sigma_U^2]$ solve

$$\text{tr}(\tilde{Q}X) = D, \quad \omega(\tilde{\zeta}^\top X \tilde{\zeta} + \sigma_U^2) = 1 - \alpha.$$

Proof. With $\tilde{H} = \tilde{\zeta}^\top$ the tilt in (5) is $(1-\alpha)\tilde{\zeta}\tilde{\zeta}^\top/(\tilde{\zeta}^\top X \tilde{\zeta} + \sigma_U^2)$, and $0 \preceq X \preceq I$ bounds ω . $\square$

Remark 10. The two scalars can be found by nested one-dimensional searches. For fixed ν , the conditions $X = \Phi(2\nu\tilde{Q} + \omega\tilde{\zeta}\tilde{\zeta}^\top)$ and $\omega(\tilde{\zeta}^\top X \tilde{\zeta} + \sigma_U^2) = 1 - \alpha$ are the optimality conditions of the strictly convex program of minimizing the weighted objective plus $\nu \text{tr}(\tilde{Q}X)$ over $0 \prec X \preceq I$, so they have exactly one solution. The function $\omega \mapsto \omega(\tilde{\zeta}^\top X \tilde{\zeta} + \sigma_U^2) - (1 - \alpha)$, with $X = \Phi(2\nu\tilde{Q} + \omega\tilde{\zeta}\tilde{\zeta}^\top)$, is continuous because Φ is, negative at $\omega = 0$ when $\alpha < 1$, and nonnegative at $\omega = (1-\alpha)/\sigma_U^2$. Any root gives that unique solution, so bisection finds it. The distortion of the penalized minimizer is nonincreasing in ν by the standard comparison of the penalized optima at two values of ν , so a second bisection matches D . No convex solver is needed.

For a holder of rank $r > 1$ the tilt is an $r \times r$ matrix that depends on the solution, and we do not have a closed form. The fixed point is nevertheless reached by iterating steps that are closed form up to a monotone one-dimensional search for the level.

Proposition 11. Let $\mathcal{P} = \{0 \prec X \preceq I, \text{tr}(\tilde{Q}X) \leq D\}$, let $X_0 = I$, and for $t \geq 0$ let X_{t+1} be the minimizer over $\mathcal{P}$ of the surrogate g_t defined in the proof. Then

$$X_{t+1} = \Phi(2\nu_t\tilde{Q} + T_t), \quad T_t = (1-\alpha)\tilde{H}^\top(\tilde{H}X_t\tilde{H}^\top + \Sigma_U)^{-1}\tilde{H},$$

where $\nu_t \geq 0$ is the multiplier of the distortion constraint, with $\nu_t = 0$ if $\text{tr}(\tilde{Q}\Phi(T_t)) \leq D$, and otherwise ν_t is a level with $\text{tr}(\tilde{Q}X_{t+1}) = D$. The weighted cost is nonincreasing along $(X_t)_{t \geq 1}$, $X_{t+1} - X_t \rightarrow 0$, and X_t converges to the frontier point of weight α .

Proof. Write the cost as $f(X) = -\frac{1}{2} \ln \det X + \frac{1-\alpha}{2} \ln \det M(X)$ with $M(X) = \tilde{H}X\tilde{H}^\top + \Sigma_U \succ 0$. The second term is concave in $X \succeq 0$, so it lies below its tangent at X_t , and

$$g_t(X) = -\frac{1}{2} \ln \det X + \frac{1}{2} \text{tr}(T_t X) + c_t, \quad c_t = \frac{1-\alpha}{2} [\ln \det M(X_t) - \text{tr}(\tilde{H}^\top M(X_t)^{-1} \tilde{H} X_t)],$$

satisfies $g_t \geq f$ on $X \succeq 0$, $g_t(X_t) = f(X_t)$, and $\nabla g_t(X_t) = \nabla f(X_t)$. The function g_t is strictly convex, so its minimizer over $\mathcal{P}$ exists and is unique (ϵI is strictly feasible and $-\ln \det$ diverges at the boundary). Its optimality conditions are $X^{-1} = 2\nu\tilde{Q} + T_t + 2\Theta$ with $\nu \geq 0$, $\Theta \succeq 0$, $\Theta(I - X) = 0$, and $\nu(\text{tr}(\tilde{Q}X) - D) = 0$. The invariant-subspace argument of Theorem 7 uses only these relations, so $X_{t+1} = \Phi(2\nu_t\tilde{Q} + T_t)$. The same argument shows that $\Phi(2\nu\tilde{Q} + T_t)$ minimizes $g_t + \nu \text{tr}(\tilde{Q}X)$ over $0 \prec X \preceq I$, so comparing the minimizers at two levels shows that its distortion is nonincreasing in ν , and it is continuous because Φ is a spectral function of the continuous map $\lambda \mapsto \min\{1, 1/\lambda\}$. Hence ν_t is as stated. The level need not be unique, but every admissible level gives the same X_{t+1} , the unique minimizer of g_t .

Monotonicity. For $t \geq 1$, $X_t \in \mathcal{P}$ by construction, so $f(X_{t+1}) \leq g_t(X_{t+1}) \leq g_t(X_t) = f(X_t)$. The start $X_0 = I$ is not feasible and serves only to define T_0 .

Compactness. Since $\min\{1, 1/\lambda\} \leq 2/(1+\lambda)$ for $\lambda \geq 0$, $\Phi(A) \preceq 2(A+I)^{-1}$, and for $A = 2\nu\tilde{Q} + T$ with $T \succeq 0$ this is at most $2(2\nu\tilde{Q} + I)^{-1}$. So $\text{tr}(\tilde{Q}\Phi(2\nu\tilde{Q} + T)) \leq \sum_i 2q_i/(2\nu q_i + 1)$, with q_i the eigenvalues of $\tilde{Q}$, uniformly in T , and $\nu_t \leq \bar{\nu}$, the level at which this bound equals D . With $T_t \preceq \bar{T} = (1-\alpha)\tilde{H}^\top \Sigma_U^{-1} \tilde{H}$, every eigenvalue of $2\nu_t\tilde{Q} + T_t$ is at most $\Lambda = \lambda_{\max}(2\bar{\nu}\tilde{Q} + \bar{T})$, so $X_{t+1} \succeq \delta I$ with $\delta = \min\{1, 1/\Lambda\}$. The iterates lie in the compact set $\mathcal{P} \cap \{X \succeq \delta I\}$.

Sufficient decrease. On $\{0 \prec X \preceq I\}$ the second derivative of $-\frac{1}{2} \ln \det X$ in a direction V is $\frac{1}{2} \text{tr}(X^{-1} V X^{-1} V) \geq \frac{1}{2} \|V\|_F^2$, so g_t is strongly convex there with modulus $\frac{1}{2}$, and the first-order optimality of X_{t+1} over the convex set $\mathcal{P}$ gives $g_t(X_t) - g_t(X_{t+1}) \geq \frac{1}{4} \|X_t - X_{t+1}\|_F^2$. Hence $f(X_t) - f(X_{t+1}) \geq \frac{1}{4} \|X_t - X_{t+1}\|_F^2$, the series of these terms is bounded by $4(f(X_1) - \min_{\mathcal{P}} f)$, and $X_{t+1} - X_t \rightarrow 0$.

Limit. Let $X_{t_k} \rightarrow \bar{X}$. Then $X_{t_k+1} \rightarrow \bar{X}$, and along a further subsequence $\nu_{t_k} \rightarrow \nu^* \in [0, \bar{\nu}]$. The matrices $\Theta_k = \frac{1}{2}(X_{t_k+1}^{-1} - 2\nu_{t_k}\tilde{Q} - T_{t_k}) \succeq 0$ satisfy $\Theta_k(I - X_{t_k+1}) = 0$, and T_t depends continuously on X_t , so Θ_k converges to some $\Theta \succeq 0$ with $\bar{X}^{-1} = 2\nu^*\tilde{Q} + T(\bar{X}) + 2\Theta$, $\Theta(I - \bar{X}) = 0$, and $\nu^*(\text{tr}(\tilde{Q}\bar{X}) - D) = 0$. These are the optimality conditions of f over $\mathcal{P}$ from the proof of Theorem 7, so $\bar{X}$ is the frontier point, which is unique, and $\nu^* > 0$ because $\nu^* = 0$ would make $\bar{X} = I$, whose distortion $D_{\max}$ exceeds D . Every limit point of the bounded sequence is that point, so the sequence converges. $\square$

Each step is a weighted reverse water-filling and a majorization–minimization step, the convex–concave procedure applied to the concave log-determinant term [7], [8], [9], [10]. Like a Blahut–Arimoto update it is explicit apart from a scalar search and decreases the cost monotonically. In 40 random instances with described variables and holders of rank between two and six in non-commuting geometry, the cost never increased beyond round-off (largest increase 2.7×10^{-15}), the distortion constraint was active at all 1023 steps (no step had $\nu_t = 0$), the relative change in the cost fell below 10^{-9} in a median of 6 and at most 21 steps, and the limit agreed with an independent determinant program to 3.3×10^{-8} in cost (one reference solve was reported as possibly inaccurate by the solver).

So for a scalar holder, the case of the scalar and coupled instances and of Appendix C, the whole frontier, partial description included, is the fixed point of reverse water-filling on a rank-one tilt of the decoder’s weight, computed as in Remark 10. As intuition for why the projection direction turns, note that a single party has $\omega = 0$ and water-fills on the fixed eigenvectors of $\tilde{Q}$, while the holder adds $\omega\tilde{\zeta}\tilde{\zeta}^\top$, and as the budget moves, the ratio ω/ν moves and the eigenvectors of $2\nu\tilde{Q} + \omega\tilde{\zeta}\tilde{\zeta}^\top$ rotate between those of $\tilde{Q}$ and the direction $\tilde{\zeta}$. In the coupled instance of Section II, the two-scalar solution agrees with the determinant program at $D = 0.3, 0.6, 0.9$, and 1.2 , with a two-dimensional description at the first three budgets and a one-dimensional one at the last.

V. EXPLICIT CASES

Use the whitened coordinates X and $\tilde{Q}$ of Section IV, and write $\tilde{J} = \Sigma_T^{1/2} J \Sigma_T^{1/2}$. Since $\Sigma_{T|S} = \Sigma_T^{1/2}(I + \tilde{J})^{-1} \Sigma_T^{1/2}$ and $\Sigma_{e0}^{-1} + J = \Sigma_T^{-1/2}(X^{-1} + \tilde{J}) \Sigma_T^{-1/2}$, the costs and the constraint of Proposition 2 become

$$r = -\frac{1}{2} \log \det X, \quad \ell = \frac{1}{2} \log \det(X^{-1} + \tilde{J}) - \frac{1}{2} \log \det(I + \tilde{J}), \quad \text{tr}(\tilde{Q}X) \leq D, \quad 0 \prec X \preceq I. \quad (6)$$

Theorem 12. Suppose $\mathbb{R}^p = \bigoplus_k \mathcal{V}_k$ is an orthogonal decomposition into subspaces invariant under both $\tilde{Q}$ and $\tilde{J}$. Then for every $\alpha \in [0, 1]$ the minimizer of Theorem 5(a), written in the coordinate X , is block-diagonal with respect to the decomposition, and the weighted program splits into one program per block, coupled only through an allocation $D = \sum_k D_k$ that equalizes the slopes of the blocks’ optimal values.

Proof. Let P_k be the orthogonal projector onto $\mathcal{V}_k$, and for signs $\epsilon_k \in \{\pm 1\}$ let $S_\epsilon = \sum_k \epsilon_k P_k$, an orthogonal symmetric matrix that commutes with $\tilde{Q}$ and $\tilde{J}$. The map $X \mapsto S_\epsilon X S_\epsilon$ preserves $\log \det X$, $\log \det(X^{-1} + \tilde{J})$ because $(S_\epsilon X S_\epsilon)^{-1} + \tilde{J} = S_\epsilon(X^{-1} + \tilde{J})S_\epsilon$, the distortion $\text{tr}(\tilde{Q}S_\epsilon X S_\epsilon) = \text{tr}(\tilde{Q}X)$, and the constraint $X \preceq I$. The minimizer is unique, so it is fixed by every S_ϵ , and then $P_k X P_l = \epsilon_k \epsilon_l P_k X P_l$ for all signs, which forces $P_k X P_l = 0$ for $k \neq l$. For block-diagonal X both costs and the distortion are sums over blocks, and the remaining problem is a separable convex program with one linear constraint, whose optimality condition is a common subgradient level. $\square$

The independent components of Section VIII are the case of two-dimensional blocks, each holding one described coordinate and its context. When the blocks are one-dimensional, which happens exactly when $\tilde{Q}$ and $\tilde{J}$ commute, the whole region is explicit.

Theorem 13. Suppose $Q\Sigma_T J = J\Sigma_T Q$, equivalently $\tilde{Q}\tilde{J} = \tilde{J}\tilde{Q}$, and let $u_1, \dots, u_p$ be a common orthonormal eigenbasis with $\tilde{Q}u_i = q_i u_i$ and $\tilde{J}u_i = j_i u_i$. For $\alpha \in [0, 1]$ the frontier point of weight α is $X^* = \sum_i x_i u_i u_i^\top$, where $x_i = 1$ if $q_i = 0$ and otherwise $x_i = \min\{1, \xi_i(\theta)\}$, with $\xi_i(\theta)$ the positive root of

$$\theta q_i j_i x^2 + (\theta q_i - \alpha j_i) x - 1 = 0 \quad (7)$$

($\xi_i = 1/(\theta q_i)$ when $j_i = 0$), and $\theta > 0$ the unique level with $\sum_i q_i x_i = D$. Its rate and leakage are

$$R = \sum_i \frac{1}{2} \log \frac{1}{x_i}, \quad L = \sum_i \frac{1}{2} \log \frac{1 + j_i x_i}{x_i(1 + j_i)}.$$

In particular the minimum leakage takes $x_i = \min\{1, (\sqrt{1 + 4j_i/(\theta q_i)} - 1)/(2j_i)\}$, read as $x_i = \min\{1, 1/(\theta q_i)\}$ when $j_i = 0$ (the limit as $j_i \rightarrow 0$), and direction i is described exactly when $\theta > (1 + \alpha j_i)/(q_i(1 + j_i))$, so the leakage-optimal description begins with the directions of largest $q_i(1 + j_i)$, the rate-optimal one with those of largest q_i .

Proof. By Theorem 12 with the one-dimensional blocks spanned by the u_i , the minimizer is $\sum_i x_i u_i u_i^\top$ and the program is $\min \sum_i \phi_i(x_i)$ subject to $\sum_i q_i x_i \leq D$ and $0 < x_i \leq 1$, with $\phi_i(x) = \frac{\alpha}{2} \ln \frac{1}{x} + \frac{1-\alpha}{2} \ln \frac{1+j_i x}{x(1+j_i)}$ in natural units. Each ϕ_i is convex and decreasing, with $-\phi'_i(x) = (1 + \alpha j_i x)/(2x(1 + j_i x))$. A direction with $q_i = 0$ does not enter the constraint, so $x_i = 1$. For the others the optimality condition is $-\phi'_i(x_i) = \frac{1}{2}\theta q_i$ when $x_i < 1$ and $-\phi'_i(1) \geq \frac{1}{2}\theta q_i$ when $x_i = 1$, for a common $\theta > 0$. The first is (7), whose roots have product $-1/(\theta q_i j_i) < 0$, so exactly one is positive, and the second reads $\theta \leq (1 + \alpha j_i)/(q_i(1 + j_i))$. Each $x_i(\theta)$ is continuous and nonincreasing, strictly decreasing where it is below one, so $\sum_i q_i x_i(\theta)$ decreases continuously and strictly from $\sum_i q_i > D$ and θ is unique. The formulas for R and L are (6) at a diagonal X . $\square$

In each direction the leakage $\frac{1}{2} \log[(1 + j_i x_i)/(x_i(1 + j_i))]$ is Steinberg's scalar Gaussian common-reconstruction function [17] at unit variance, distortion x_i , and squared correlation $j_i/(1 + j_i)$ between the direction and the holder's observation. The theorem is the allocation of the distortion budget among such problems, and the joint frontier that goes with it.

Corollary 14. In commuting geometry every optimal description, at every budget and weight, reads the fixed directions u_i , consistent with Theorem 20. The rate-optimal and leakage-optimal descriptions nonetheless differ whenever two directions described by the rate-optimal one have different ratios j_i/q_i .

Proof. The first statement is Theorem 13. If the two optimal descriptions coincided, they would describe the same directions with the same x_i . For those directions $x_i = 1/(\theta_R q_i)$ and $x_i(1 + j_i x_i) = 1/(\theta_L q_i)$, so $1 + j_i x_i = \theta_R/\theta_L$ is the same for all of them, and $j_i x_i = j_i/(\theta_R q_i)$ forces j_i/q_i to be the same. $\square$

When the decoder reproduces the whole source, the same per-direction formula holds without commuting geometry at every small budget, in the basis of the pencil (J, Q) instead of the common eigenbasis.

Theorem 15. Suppose $Q \succ 0$, and let $\Pi = Q^{-1/2} J Q^{-1/2} = \sum_i \pi_i w_i w_i^\top$ with orthonormal w_i . For $\alpha \in [0, 1]$ and $\theta > 0$ let $\xi_i(\theta)$ be the positive root of

$$\theta \pi_i \xi^2 + (\theta - \alpha \pi_i) \xi - 1 = 0 \quad (8)$$

($\xi_i = 1/\theta$ when $\pi_i = 0$), let $\theta(D)$ be the unique level with $\sum_i \xi_i = D$, and let $\Sigma_{e0}(D) = Q^{-1/2} (\sum_i \xi_i w_i w_i^\top) Q^{-1/2}$. Then $\Sigma_{e0}(D)$ is increasing in D in the Löwner order, the set of D with $\Sigma_{e0}(D) \preceq \Sigma_T$ is an interval $(0, D_{\text{int}}(\alpha)]$ with $D_{\text{int}}(\alpha) > 0$, and on it $\Sigma_{e0}(D)$ is the frontier point of weight α , with

$$R = \frac{1}{2} \log \det(Q\Sigma_T) - \sum_i \frac{1}{2} \log \xi_i, \quad L = \frac{1}{2} \log \det(Q\Sigma_{T|S}) + \sum_i \frac{1}{2} \log \frac{1 + \pi_i \xi_i}{\xi_i}.$$

No commuting assumption is made, and whether a budget lies in the interval is a single test of the Löwner order.

Proof. In the coordinate $\Xi = Q^{1/2} \Sigma_{e0} Q^{1/2}$ the distortion is $\text{tr } \Xi$, and $\Sigma_e^{-1} = Q^{1/2} (\Xi^{-1} + \Pi) Q^{1/2}$, which gives the formulas for R and L at $\Xi = \sum_i \xi_i w_i w_i^\top$. When the constraint $\Sigma_{e0} \preceq \Sigma_T$ is slack, its multiplier vanishes, and the stationarity condition in the proof of Proposition 6, multiplied on the right by Σ_{e0} , reads $\alpha I + (1 - \alpha)(I + J\Sigma_{e0})^{-1} = 2\nu Q\Sigma_{e0}$, which in the coordinate Ξ is

$$\alpha I + (1 - \alpha)(I + \Pi\Xi)^{-1} = \theta\Xi, \quad \theta = 2\nu.$$

For $\alpha < 1$ the right side is symmetric, so $I + \Pi\Xi$ is symmetric and $\Pi\Xi = \Xi\Pi$. For $\alpha = 1$ the condition is $\Xi = I/\theta$. Either way Ξ leaves each eigenspace of Π invariant, and on the eigenspace of π_i every eigenvalue ξ of Ξ satisfies $\alpha + (1 - \alpha)/(1 + \pi_i\xi) = \theta\xi$, which is (8). Its left side decreases and its right side increases in $\xi > 0$, so the positive root is unique and decreases in θ . Hence Ξ is a multiple of the identity on each eigenspace of Π and diagonal in any eigenbasis w_i , $\theta(D)$ is unique, each ξ_i increases in D , and so does $\Sigma_{e0}(D)$, by congruence. As $D \rightarrow 0$, $\Sigma_{e0}(D) \rightarrow 0 \prec \Sigma_T$. When $\Sigma_{e0}(D) \preceq \Sigma_T$, the point satisfies the Karush–Kuhn–Tucker conditions with zero multiplier for the cap, which suffice by convexity, and Theorem 5(a) makes it the unique minimizer. $\square$

When $Q\Sigma_T J = J\Sigma_T Q$ and no cap of Theorem 13 is active, the two results agree, with $\xi_i = q_i x_i$ and $\pi_i = j_i/q_i$. When Q is singular, Proposition 6 shows that the cap is always active, and the error covariance equals Σ_T on a subspace of dimension at least $p - \text{rank } Q$.

VI. THE ONE-DIMENSIONAL CASE

Write $\Delta = D_{\max} - D$ for the reduction in distortion the description must buy, and

$$A(\Delta) = \Sigma_T Q \Sigma_T - \Delta \Sigma_T.$$

A one-dimensional description is $Z = b^\top T + N$ with $b \neq 0$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, reproduced by $\mathbb{E}[Y | Z]$. Its projection direction is b , and the direction in the space of the described variable along which it reduces the error is $F\Sigma_T b$.

Theorem 16. (a) A one-dimensional description meets the distortion constraint with finite leakage if and only if $\Delta < \lambda_{\max}(W^{1/2}\Sigma_Y W^{1/2})$, where $\Sigma_Y = F\Sigma_T F^\top$, and the least leakage among such descriptions is

$$L_{1d}(D) = \frac{1}{2} \log \left(1 + \frac{\Delta}{\mu_{\max}(\Delta)} \right), \quad (9)$$

where $\mu_{\max}(\Delta) > 0$ is the largest root of $\det(A(\Delta) - \mu\Sigma_{T|S}) = 0$. It is attained with b a corresponding generalized eigenvector and $\sigma^2 = b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$, at which the distortion constraint is active.

(b) Suppose $\Delta < \lambda_{\max}(W^{1/2}\Sigma_Y W^{1/2})$ and that $\mu_{\max}(\Delta)$ is simple (if it is not, two distinct one-dimensional optima exist, their midpoint in the coordinate Σ_{e0} is strictly better by Theorem 5, and $L(D) < L_{1d}(D)$). Let Σ_{e0} and Σ_e be the error covariances of the description of (a), and let $\Gamma(\varpi) = \frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0}$. Then $L(D) = L_{1d}(D)$ if and only if there is $\varpi \geq 0$ with $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and then $\varpi = \frac{1}{2}b^\top \Sigma_e b / (b^\top \Sigma_{e0} Q \Sigma_{e0} b)$.

(c) If the reproduction is scalar ($m = 1$), then $L(D) = L_{1d}(D)$ for every $0 < D < D_{\max}$.

Proof. (a) For the description Z , $\text{Cov}(T | Z) = \Sigma_T - \Sigma_T b b^\top \Sigma_T / (b^\top \Sigma_T b + \sigma^2)$, so the distortion of $\mathbb{E}[Y | Z]$ is $D_{\max} - b^\top \Sigma_T Q \Sigma_T b / (b^\top \Sigma_T b + \sigma^2)$, and the constraint reads $\sigma^2 \leq b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$. The leakage $I(T; Z | S) = \frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b / \sigma^2)$ decreases in σ^2 , so at the largest admissible σ^2 , where the constraint is active, it equals $\frac{1}{2} \log(1 + \Delta b^\top \Sigma_{T|S} b / b^\top A(\Delta) b)$, finite exactly when $b^\top A(\Delta) b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A(\Delta) b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is $\mu_{\max}$. Some b has $b^\top A(\Delta) b > 0$ if and only if $\Sigma_T^{1/2} Q \Sigma_T^{1/2} \not\preceq \Delta I$, and the nonzero eigenvalues of that matrix are those of $W^{1/2}\Sigma_Y W^{1/2}$.

(b) The description of (a) has an active constraint and $\Sigma_e^{-1} = \Sigma_{T|S}^{-1} + b b^\top / \sigma^2$, so $\Sigma_{T|S} - \Sigma_e$ is a positive multiple of $\Sigma_{T|S} b b^\top \Sigma_{T|S}$ and $\Sigma_e^{-1} \Sigma_{T|S} b$ is a positive multiple of b . With $\Theta = \Sigma_e^{-1} \Gamma(\varpi) \Sigma_e^{-1}$, the conditions (4) at this point read $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and Theorem 5(c) makes them necessary and sufficient. Taking $b^\top \Gamma(\varpi) b = 0$ gives ϖ .

(c) By Lemma 1 the optimum is attained by a jointly Gaussian channel $\hat{Y} = a^\top T + N$. If $\text{Cov}(Y, \hat{Y}) = 0$ its distortion is at least $D_{\max} > D$. Otherwise $\mathbb{E}[Y | \hat{Y}]$ is a nonzero multiple of $\hat{Y}$, which leaves the leakage unchanged and does not raise the distortion, so the optimum is a one-dimensional description. $\square$

Corollary 17. Let $T = (Y, V)$ be scalar with unit variances and correlation ρ , $|\rho| < 1$, let $W = 1$, and let $S = V + U$ with $\text{Var } U = \tau^2 > 0$, and write $s = 1 + \tau^2$. For $0 < D < 1$,

$$L(D) = \frac{1}{2} \log g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds},$$

the larger root of $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$, attained by $\hat{Y} = aY + bV + N$ with $a = (g^* - 1)/g^*$, $b = (g^* - 1)\rho/(g^*(g^*s - 1))$, and the noise variance that makes the distortion equal to D .

Proof. By Theorem 16(c), $L(D) = \frac{1}{2} \log(1 + (1 - D)/\mu_{\max})$ with $\Delta = 1 - D$. Substituting $\mu = (1 - D)/(g - 1)$ into $\det(A(1 - D) - \mu\Sigma_{T|S})$, with $\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $Q = e_1 e_1^\top$, and $J = e_2 e_2^\top / \tau^2$, gives $-(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a direct computation confirmed by computer algebra (Appendix D). Since $P(1) = (D - 1)\tau^2 < 0$ and $P(0) = 1 - \rho^2 > 0$, P has one root in $(0, 1)$, which corresponds to a negative μ , and one root $g^* > 1$, which corresponds to the unique positive root $\mu_{\max}$. The projection direction is the corresponding generalized eigenvector. Normalizing the reproduction gives the stated coefficients. $\square$

For a scalar described variable the whole frontier is explicit up to one scalar root, with any context and any holder.

Theorem 18. Let $m = 1$, write $Q = qq^\top$, and for $\varrho \in [0, 1]$ let

$$\Sigma_\varrho = (1 - \varrho)\Sigma_T + \varrho\Sigma_{T|S}, \quad b(\varrho) = \frac{\Sigma_\varrho^{-1}\Sigma_T q}{q^\top \Sigma_T \Sigma_\varrho^{-1} \Sigma_T q},$$

$u(\varrho) = b^\top \Sigma_T b$, and $v(\varrho) = b^\top (\Sigma_T - \Sigma_{T|S})b$. For $\alpha \in [0, 1]$ and $0 < D < D_{\max}$, the frontier point of weight α , the minimizer of $\alpha R + (1 - \alpha)L$, is the one-dimensional description $b(\varrho)^\top T + N$ with $\text{Var } N = (1 - \Delta u)/\Delta$, where ϱ is a root of

$$\varrho(1 - \Delta v(\varrho)) = (1 - \alpha)(1 - \Delta u(\varrho)) \quad (10)$$

in $[0, 1 - \alpha]$, the one of least weighted cost. Its rate and leakage are

$$R = -\frac{1}{2} \log(1 - \Delta u), \quad L = R + \frac{1}{2} \log(1 - \Delta v) = \frac{1}{2} \log \frac{1 - \alpha}{\varrho} \quad (\alpha < 1).$$

The roots are finitely many, every root is a feasible description, and at $\alpha = 0$ the root is unique. At $\alpha = 1$ the root is $\varrho = 0$.

Proof. By Proposition 6 the minimizer is a one-dimensional description $b^\top T + N$ (rank zero cannot reduce the distortion), at the active noise level by Theorem 5(a). A direction with $q^\top \Sigma_T b = 0$ reduces no distortion, so normalize $q^\top \Sigma_T b = 1$. From the proof of Theorem 16(a) the active noise variance is $\sigma^2 = b^\top A(\Delta)b/\Delta = (1 - \Delta u)/\Delta$, so the rate is $\frac{1}{2} \log(1 + u/\sigma^2) = -\frac{1}{2} \log(1 - \Delta u)$, and the leakage is $\frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b/\sigma^2) = R + \frac{1}{2} \log(1 - \Delta v)$, since $b^\top \Sigma_{T|S} b = u - v$. The weighted cost $-\frac{1}{2} \ln(1 - \Delta u) + \frac{1 - \alpha}{2} \ln(1 - \Delta v)$ is smooth on the open set where $1 - \Delta u > 0$, the minimizer lies in it, and stationarity on the hyperplane $q^\top \Sigma_T b = 1$ gives $\Sigma_T b - \varrho(\Sigma_T - \Sigma_{T|S})b \propto \Sigma_T q$ with $\varrho = (1 - \alpha)(1 - \Delta u)/(1 - \Delta v)$, which is (10) and $b = b(\varrho)$, with $\Sigma_\varrho \succ 0$. Since $\Sigma_{T|S} \succ 0$, $u - v = b^\top \Sigma_{T|S} b > 0$. So for $\alpha < 1$ a root has $1 - \Delta v > 1 - \Delta u$ and both factors positive, hence $0 < \varrho < 1 - \alpha$, and every root is a feasible description, while $\alpha = 1$ forces $\varrho = 0$. The equation is rational in ϱ and not identically zero, since at $\varrho = 0$ its right side is $(1 - \alpha)D/D_{\max} > 0$. So the roots are finitely many, and distinct roots give distinct descriptions, so the root of least cost is the unique minimizer. At a root, $(1 - \Delta v)/(1 - \Delta u) = (1 - \alpha)/\varrho$, which gives the last form of L .

For $\alpha = 0$, put $\mu = \Delta\varrho/(1 - \varrho)$, so $\Sigma_\varrho \propto \Delta\Sigma_T + \mu\Sigma_{T|S}$ and $(\Delta\Sigma_T + \mu\Sigma_{T|S})b = c\Sigma_T q$ for a scalar c . Multiplying by $b^\top$ gives $c = \Delta u + \mu(u - v)$, and (10) with $\alpha = 0$ reduces to $\Delta u + \mu(u - v) = 1$, that is, $c = 1$. Multiplying $b = c(\Delta\Sigma_T + \mu\Sigma_{T|S})^{-1}\Sigma_T q$ by $q^\top \Sigma_T$ shows that $c = 1$ is the secular equation (11) below, whose positive root is unique. $\square$

So every optimal description of a scalar variable, at every budget and weight, reads a direction on one explicit curve, from $b(0) \propto q$, which describes Y itself, toward $b(1) \propto \Sigma_{T|S}^{-1}\Sigma_T q = q + J\Sigma_T q$. The rate-optimal description has $\varrho = 0$ at every budget. The leakage-optimal one moves along the curve with the budget, with $\varrho \rightarrow 1$ as $D \rightarrow D_{\max}$, where it reads $q + J\Sigma_T q$, and $\varrho \rightarrow 0$ as $D \rightarrow 0$, where it must reveal Y . At $\alpha = 0$ the stationarity condition is the pencil of Theorem 16. With $Q = qq^\top$ the eigenvector equation $A(\Delta)b = \mu\Sigma_{T|S}b$ reads $(\Delta\Sigma_T + \mu\Sigma_{T|S})b \propto \Sigma_T q$, and with $q^\top \Sigma_T b = 1$ it gives $1 - \Delta u = \mu(u - v)$ and $1 - \Delta v = (\mu + \Delta)(u - v)$, so $\varrho = \mu_{\max}/(\Delta + \mu_{\max})$ and $L = \frac{1}{2} \log(1 + \Delta/\mu_{\max})$. The root is the unique positive solution of the secular equation

$$q^\top \Sigma_T (\Delta\Sigma_T + \mu\Sigma_{T|S})^{-1} \Sigma_T q = 1, \quad (11)$$

whose left side decreases from $D_{\max}/\Delta > 1$ to 0 on $\mu > 0$. In the basis of the pencil $(\Sigma_T, \Sigma_{T|S})$ it is a rational equation with p poles, and for $p = 2$ it is the quadratic of Corollary 17.

VII. DIRECTIONS AND MISALIGNMENT

Proposition 19. *With no holder ($J = 0$), the best one-dimensional description reads the top generalized eigenvector of the pencil $(\Sigma_T Q \Sigma_T, \Sigma_T)$ at every budget, and the rate-optimal description reverse water-fills on the eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ along fixed directions.*

Proof. With $J = 0$, $\Sigma_{T|S} = \Sigma_T$ and $A(\Delta)b = \mu \Sigma_T b$ reads $\Sigma_T Q \Sigma_T b = (\mu + \Delta) \Sigma_T b$, whose eigenvectors do not depend on Δ . The rate-optimal description is the weighted quadratic rate-distortion solution [20]. In the coordinates $W^{1/2} Y$ the distortion is squared error, and reverse water-filling describes fixed principal components [3]. $\square$

The Gaussian information bottleneck has the same property. Its optimal projections are fixed eigenvectors, and its tradeoff parameter sets only their number and scale [4]. The leakage-optimal description behaves differently.

Theorem 20. *Let $\mu_{\max}(\Delta)$ be simple and positive on an open interval $\mathcal{I} \subseteq (0, \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2}))$, with generalized eigenvector $b(\Delta)$. Then $b(\Delta)$ spans the same line for every $\Delta \in \mathcal{I}$ if and only if, for some $\Delta_0 \in \mathcal{I}$, $b_0 = b(\Delta_0)$ satisfies*

$$Q \Sigma_T b_0 = \lambda b_0 \quad \text{and} \quad J \Sigma_T b_0 = \varsigma b_0 \quad (12)$$

for scalars $\lambda > 0$ and $\varsigma \geq 0$, that is, if and only if b_0 is a common generalized eigenvector of the pencils $(\Sigma_T Q \Sigma_T, \Sigma_T)$ and $(\Sigma_T J \Sigma_T, \Sigma_T)$. Otherwise $b(\Delta)$ is constant on no open subinterval of $\mathcal{I}$.

Proof. A simple eigenvalue of a symmetric pencil with positive definite second matrix, and its eigenvector, are analytic in the parameter [26]. Write $A(\Delta) = A_0 - \Delta \Sigma_T$.

Suppose b is constant on an open subinterval. Differentiating $(A_0 - \Delta \Sigma_T)b = \mu_{\max}(\Delta) \Sigma_{T|S} b$ gives $\Sigma_T b = c \Sigma_{T|S} b$ for a scalar c , and $c = b^\top \Sigma_T b / b^\top \Sigma_{T|S} b \geq 1$ because $\Sigma_{T|S} \preceq \Sigma_T$. Then $A_0 b = (\Delta + \mu_{\max}/c) \Sigma_T b$, that is, $Q \Sigma_T b = \lambda b$ with $\lambda = \Delta + \mu_{\max}/c > 0$. Multiplying $\Sigma_T b = c \Sigma_{T|S} b$ by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J \Sigma_T b = (c - 1)b$, so $\varsigma = c - 1 \geq 0$. This proves the last sentence and the forward direction.

Conversely, if b_0 satisfies (12), then $\Sigma_{T|S}^{-1} \Sigma_T b_0 = (1 + \varsigma) b_0$, so $\Sigma_T b_0 = (1 + \varsigma) \Sigma_{T|S} b_0$ and $A(\Delta) b_0 = (\lambda - \Delta) \Sigma_T b_0 = (\lambda - \Delta)(1 + \varsigma) \Sigma_{T|S} b_0$ for every Δ , so b_0 is a generalized eigenvector with eigenvalue $(\lambda - \Delta)(1 + \varsigma)$ throughout. Let $\mathcal{E} = \{\Delta \in \mathcal{I} : (\lambda - \Delta)(1 + \varsigma) = \mu_{\max}(\Delta)\}$. It contains Δ_0 and is closed in $\mathcal{I}$ by continuity. It is open, because near a point of $\mathcal{E}$ the simple eigenvalue $\mu_{\max}$ is the only eigenvalue of the pencil close to it, and $(\lambda - \Delta)(1 + \varsigma)$ is an eigenvalue close to it. Since $\mathcal{I}$ is connected, $\mathcal{E} = \mathcal{I}$, and by simplicity $b(\Delta)$ spans the line of b_0 throughout. $\square$

Corollary 21. *Suppose the holder observes only the context, that is, in coordinates $T = (Y, V)$, $F = [I \ 0]$ and $H = [0 \ H_V]$. Then (12) holds if and only if $b_0 = F^\top \beta$ with $W \Sigma_Y \beta = \lambda \beta$ and $\text{Cov}(S, \beta^\top Y) = 0$.*

Proof. $Q \Sigma_T b_0 = \lambda b_0$ with $\lambda > 0$ puts b_0 in the range of Q , which is the range of $F^\top$, so $b_0 = F^\top \beta$ and $F^\top W \Sigma_Y \beta = \lambda F^\top \beta$. The vector $J \Sigma_T b_0$ lies in the range of $H^\top$, which meets the range of $F^\top$ only at zero, so $J \Sigma_T b_0 = \varsigma b_0$ forces $J \Sigma_T b_0 = 0$. With $w = \Sigma_T b_0$, $w^\top J w = 0$ gives $\Sigma_U^{-1/2} H w = 0$, that is, $\Sigma_U^{-1} H \Sigma_T F^\top \beta = 0$, which is $\text{Cov}(S, \beta^\top Y) = 0$. The converse is direct. $\square$

The same condition decides whether the two optimal descriptions coincide. Let b_R be the top generalized eigenvector of $(\Sigma_T Q \Sigma_T, \Sigma_T)$, the projection direction of the best one-dimensional rate description.

Theorem 22. *Let $\lambda_1 \geq \lambda_2$ be the two largest eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$, with $\lambda_2 = 0$ when $m = 1$, and suppose $\Delta \leq \lambda_1 - \lambda_2$, the range in which the rate-optimal description is one-dimensional, $b_R^\top T + N$. If $J \Sigma_T b_R$ is not parallel to b_R , the rate-optimal description is not leakage-optimal and the leakage-optimal description is not rate-optimal, so each pays a strictly positive excess in the other cost. If $J \Sigma_T b_R$ is parallel to b_R , then b_R is a generalized eigenvector of the leakage pencil at every Δ , and inside the one-dimensional range of Theorem 16 the two descriptions coincide exactly when it is the top one.*

Proof. In the stated range reverse water-filling describes only the principal component, so the rate-optimal description is the one-dimensional description reading b_R , where $Q \Sigma_T b_R = \lambda_1 b_R$. If it were leakage-optimal,

then $L(D) = L_{1d}(D)$ and b_R would maximize the Rayleigh quotient of Theorem 16(a), hence be a generalized eigenvector of $(A(\Delta), \Sigma_{T|S})$, that is, $(\lambda_1 - \Delta)\Sigma_T b_R \propto \Sigma_{T|S} b_R$. Multiplying by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J\Sigma_T b_R \propto b_R$. So when $J\Sigma_T b_R$ is not parallel to b_R the rate-optimal description is not leakage-optimal, and since both minimizers of Theorem 5(a) are unique, the leakage-optimal description is not rate-optimal either. The same computation read backward shows that $J\Sigma_T b_R \propto b_R$ makes b_R a generalized eigenvector at every Δ , and inside the one-dimensional range the leakage-optimal description reads the top one. $\square$

In the scalar case of Corollary 17, $b_R = e_1$ and $J\Sigma_T e_1 = (\rho/\tau^2)e_2$, which is parallel to e_1 only when $\rho = 0$. So the two descriptions differ exactly when the context is correlated with the described variable. The rate-optimal channel is the classical reverse channel, with leakage $\frac{1}{2} \log[((1-D)(1-\rho^2/s) + D)/D]$, strictly above $L(D)$, and the leakage-optimal channel of Corollary 17 has rate strictly above $\frac{1}{2} \log(1/D)$. The rate excess is unbounded toward the clean-context boundary $\tau^2 \rightarrow 0$ at $D = 1 - \rho^2$, where the leakage-optimal channel pins the reproduction to the context.

Corollary 23. *There are sources for which the leakage-optimal one-dimensional projection takes different directions at two budgets. For such a source no description problem for a single party with a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at both budgets.*

Proof. Take the scalar source of Corollary 17 with $\rho \neq 0$. By Theorem 16(c) the leakage-optimal description is the one-dimensional description at every D , and its projection direction is proportional to $(g^*s - 1, \rho)$, the ratio of the coefficients in Corollary 17. Since $L(D)$ is strictly decreasing in D (if $L(D_1) = L(D_2)$ with $D_1 < D_2$, the unique minimizer at D_2 would be the one at D_1 , whose constraint is slack at D_2 , contradicting Theorem 5(a)), g^* is strictly decreasing in D , so the projection line differs at any two budgets. By Proposition 19, applied to any fixed weight on T , the best one-dimensional description for a single party reads the same line at both budgets. The coupled instance of Section II, whose leakage-optimal descriptions reduce the error along 47.48° and 41.15° at two budgets, illustrates the same effect in the space of the described variable. $\square$

A description whose projection direction is fixed, such as one designed for a single party with a weighted quadratic distortion, therefore loses something at all but isolated budgets. The loss has an explicit lower bound. Let $\mu_1 = \mu_{\max} \geq \mu_2 \geq \dots$ be the roots of $\det(A(\Delta) - \mu\Sigma_{T|S}) = 0$, and measure angles between directions in the inner product of $\Sigma_{T|S}$, which does not depend on the budget.

Proposition 24. *Let $0 < \Delta < \lambda_{\max}(W^{1/2}\Sigma_Y W^{1/2})$, so that $\mu_1 > 0$, and let $b(\Delta)$ be a generalized eigenvector for μ_1 . For b with $\mu_b = b^T A(\Delta) b / b^T \Sigma_{T|S} b > 0$, the one-dimensional description with projection direction b at the active noise level has leakage $\ell_b(\Delta) = \frac{1}{2} \log(1 + \Delta/\mu_b)$, and with $a = (\mu_1 - \mu_2) \sin^2 \psi$, where ψ is the angle between b and $b(\Delta)$, we have $a < \mu_1$ and*

$$\ell_b(\Delta) - L_{1d}(D) \geq \frac{1}{2} \log\left(1 + \frac{\Delta a}{(\mu_1 - a)(\mu_1 + \Delta)}\right), \quad (13)$$

with equality when $p = 2$.

Proof. The leakage is the one computed in the proof of Theorem 16(a), where $\mu_1 > 0$ is also shown. In a $\Sigma_{T|S}$ -orthonormal basis of generalized eigenvectors v_i , with $v_1 = b(\Delta)/\|b(\Delta)\|_{\Sigma_{T|S}}$, write $b = \sum_i c_i v_i$. Then $\mu_b = \sum_i \mu_i c_i^2 / \sum_i c_i^2$, so $\mu_1 - \mu_b \geq (\mu_1 - \mu_2) \sum_{i \geq 2} c_i^2 / \sum_i c_i^2 = a$, with equality when $p = 2$, and $a \leq \mu_1 - \mu_b < \mu_1$. Finally $(1 + \Delta/\mu_b)/(1 + \Delta/\mu_1) = 1 + \Delta x / ((\mu_1 - x)(\mu_1 + \Delta))$ with $x = \mu_1 - \mu_b$, which increases in $x \in [a, \mu_1]$. $\square$

Theorem 25. *Let Δ_1 and Δ_2 satisfy the hypotheses of Proposition 24, and let ψ_{12} be the angle between $b(\Delta_1)$ and $b(\Delta_2)$. Every one-dimensional description with the same projection direction at both budgets has, at one of them, excess leakage over L_{1d} at least*

$$\min_{i=1,2} \frac{1}{2} \log\left(1 + \frac{\Delta_i a_i}{(\mu_1 - a_i)(\mu_1 + \Delta_i)}\right), \quad a_i = (\mu_1 - \mu_2) \sin^2(\psi_{12}/2),$$

with μ_1 and μ_2 evaluated at Δ_i , and a term read as $+\infty$ when $a_i \geq \mu_1$. If $\mu_{\max}$ is simple on an open interval containing Δ_1 , then unless $b(\Delta)$ is constant there (Theorem 20), $\psi_{12} > 0$ for all but isolated Δ_2 in that interval, by analyticity.

Proof. The angle between lines is a metric on the projective space of the $\Sigma_{T|S}$ inner product, so the angles from b to $b(\Delta_1)$ and to $b(\Delta_2)$ sum to at least ψ_{12} , and one of them is at least $\psi_{12}/2$. At that budget, a direction whose angle gives $a \geq \mu_1$ has $\mu_b \leq 0$ and infinite leakage, and otherwise the bound (13) increases with $\sin^2 \psi$ on $[0, \pi/2]$. $\square$

Since $L(D) \leq L_{1d}(D)$, the bounds are also lower bounds on the excess over the least leakage of any description, and they bound that excess exactly where $L(D) = L_{1d}(D)$, at every budget for a scalar described variable and inside the certificate of Theorem 16(b) in general. The bounds need no search. The two-budget values below were found by search over directions. In the coupled instance the rate-optimal description, whose projection direction is fixed, pays 0.139 bits of excess leakage at $D = 1.1$ (bound 0.098) and 0.011 bits at $D = 1.38$ (bound 0.010). For the two budgets $D = 1.38$ and 1.1, whose optimal projection directions are 24.8° apart in the $\Sigma_{T|S}$ metric, the best single fixed direction pays only 0.0003 bits at the worse of them. In this instance the excess comes from the choice of direction, and little from holding it fixed. A less noisy context changes this. For the scalar source with $(\rho^2, \tau^2) = (0.75, 0.05)$ and budgets $D = 0.1$ and 0.7, the optimal directions are 49.8° apart in the $\Sigma_{T|S}$ metric, and every fixed direction pays at least 0.0245 bits at one budget (bound 0.0048), more than the least leakage 0.0216 bits at $D = 0.7$. With $(\rho^2, \tau^2) = (0.5, 10^{-3})$ and $D = 0.2$ and 0.8, the directions are 84.0° apart and every fixed direction pays at least 0.0386 bits (bound 0.0004), against a least leakage of 0.0005 bits at $D = 0.8$.

VIII. INDEPENDENT COMPONENTS

Suppose $Y = (Y_1, \dots, Y_m)$, W is diagonal with entries w_j , and each Y_j has its own context V_j and holder observation $S_j = V_j + U_j$, the triples (Y_j, V_j, S_j) being independent, with $\text{Var } Y_j = \sigma_j^2$, $\text{Var } V_j = 1$, $\mathbb{E}[Y_j V_j] = \rho_j \sigma_j$, $|\rho_j| < 1$, and $\text{Var } U_j = \tau_j^2 > 0$. Absorb w_j into σ_j^2 . Write ℓ_j for the scalar function $\delta \mapsto \frac{1}{2} \log g^*(\delta)$ of Corollary 17 at (ρ_j, τ_j^2) , extended by zero for $\delta \geq 1$, and $L_j(D_j) = \ell_j(D_j/\sigma_j^2)$.

Lemma 26. For $0 < \delta < 1$, with $g = g^*(\delta)$, $s = 1 + \tau^2$, $k = gs - 1$, and $\Delta_\delta = \sqrt{(\delta + s - \rho^2)^2 - 4\delta s(1 - \rho^2)} > 0$,

$$\ell'(\delta) = -\frac{k}{2 \ln 2 \Delta_\delta}, \quad \ell''(\delta) = \frac{(1 - \rho^2)k^3 + \rho^2 \tau^2 g^2 (\tau^4 g + \tau^2 (2g + 1) + g + 1)}{2 \ln 2 g \Delta_\delta^3} > 0,$$

so ℓ is strictly convex and decreasing, $\ell' \rightarrow -\infty$ as $\delta \rightarrow 0$, and $\ell' \rightarrow -\tau^2/(2 \ln 2(\rho^2 + \tau^2))$ as $\delta \rightarrow 1$.

Proof. The larger root is simple, with $\partial_g P = \Delta_\delta$ there, so g^* is smooth. Differentiating $P(g^*(\delta)) = 0$ with $\partial_\delta P = gk$ gives $g' = -gk/\Delta_\delta$ and the first formula. Differentiating once more and eliminating δ through $\delta gk = (s - \rho^2)g - (1 - \rho^2)$ gives the second, an identity of polynomials confirmed by computer algebra (Appendix D). Its numerator is positive. The limits follow from $g^* \rightarrow \infty$, $\Delta_\delta \rightarrow s - \rho^2$ as $\delta \rightarrow 0$ and $g^* \rightarrow 1$, $k \rightarrow \tau^2$, $\Delta_\delta \rightarrow \rho^2 + \tau^2$ as $\delta \rightarrow 1$. $\square$

Define the activation level $\kappa_j = -L'_j(\sigma_j^2 -) = \tau_j^2/(2 \ln 2 \sigma_j^2(\rho_j^2 + \tau_j^2))$. For the rate it would be $1/(2 \ln 2 \sigma_j^2)$.

These components are two-dimensional blocks of Theorem 12, each solved by Corollary 17.

Theorem 27. (a) $\mathcal{R}(D)$ is the union, over $D_j \geq 0$ with $\sum_j D_j \leq D$, of the Minkowski sums of the scalar regions of the components, attained by product channels. (b) $L(D) = \min \sum_j L_j(D_j)$ over such allocations, with a unique minimizer $D_j = D_j(\vartheta^*)$, where $D_j(\vartheta) = \sigma_j^2$ for $\vartheta \leq \kappa_j$ and otherwise solves $-L'_j(D_j) = \vartheta$, and ϑ^* is the unique level with $\sum_j D_j(\vartheta^*) = D$. (c) As D decreases, the leakage-optimal description begins to describe component j in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$, while the rate-optimal description proceeds in decreasing order of σ_j^2 .

Proof. (a) By Theorem 3 it suffices to identify the single-letter union. For an admissible channel with $D_j = \mathbb{E}(Y_j - \hat{Y}_j)^2$, each induced channel $p(\hat{y}_j | t_j)$ satisfies $\hat{Y}_j - T_j - S_j$, and since the pairs (T_j, S_j) are independent the chain rule gives $I(T; \hat{\mathbf{Y}}) \geq \sum_j I(T_j; \hat{Y}_j)$ and $I(T; \hat{\mathbf{Y}} | S) = \sum_j [I(T_j; \hat{\mathbf{Y}}, S, T^{j-1}) - I(T_j; S_j)] \geq \sum_j I(T_j; \hat{Y}_j | S_j)$. A product of component channels attains the sum with equality. (b) Each L_j is convex and nonincreasing, strictly convex on $(0, \sigma_j^2)$ by Lemma 26, with left derivative $-\kappa_j$ at σ_j^2 . An allocation is optimal if and only if some $\vartheta \geq 0$ lies in $-\partial L_j(D_j)$ for every j with $\vartheta(\sum_j D_j - D) = 0$. Since $D < \sum_j \sigma_j^2$, $\vartheta > 0$. No D_j is 0 because $-L'_j \rightarrow \infty$ there. The sum $\sum_j D_j(\vartheta)$ decreases continuously and strictly from $\sum_j \sigma_j^2$ at $\min_j \kappa_j$ to 0, so ϑ^* is unique. (c) $D_j(\vartheta) < \sigma_j^2$ exactly when $\vartheta > \kappa_j$, and ϑ^* increases as D decreases. $\square$

Corollary 28. *If $m = 2$, $\sigma_1^2 > \sigma_2^2$, and $\sigma_1^2(1 + \rho_1^2/\tau_1^2) < \sigma_2^2(1 + \rho_2^2/\tau_2^2)$, then for every $D \in [\max\{2\sigma_2^2, \sigma_1^2 + D_2(\kappa_1)\}, \sigma_1^2 + \sigma_2^2)$ the rate-optimal description describes component 1 alone and the leakage-optimal description describes component 2 alone.*

Proof. Reverse water-filling describes component 1 alone exactly for $2\sigma_2^2 \leq D < \sigma_1^2 + \sigma_2^2$. Theorem 27(b) describes component 2 alone exactly when $\kappa_2 < \vartheta^* \leq \kappa_1$, that is, for $\sigma_1^2 + D_2(\kappa_1) \leq D < \sigma_1^2 + \sigma_2^2$. $\square$

IX. DECODER SIDE INFORMATION

Suppose the decoder also observes $C = H_C T + N_C$, with $N_C \sim \mathcal{N}(0, \Sigma_{N_C})$, $\Sigma_{N_C} \succ 0$, independent of T , U , and all coding randomness, and reconstructs $\hat{Y}^n = g_n(M, C^n)$. The encoder does not see C . Rate, distortion, and leakage $L_n = \frac{1}{n} H(M | S^n)$ are as in Section II. Write $J_S := J$ for the holder's information, $J_C := H_C^T \Sigma_{N_C}^{-1} H_C$, and $\Sigma_{T|C} = (\Sigma_T^{-1} + J_C)^{-1}$.

Theorem 29. *Let $\mathcal{R}_C(D)$ be the set of pairs (R, L) for which there are random variables V_0, V with values in Euclidean spaces, $V_0 - V - T - (C, S)$, and $I(T; V) < \infty$, and a measurable $\hat{y}$ with $\mathbb{E}[(Y - \hat{y}(V, C))^T W (Y - \hat{y}(V, C))] \leq D$, such that*

$$R \geq I(T; V | C), \quad L \geq I(T; V | C) + I(V_0; C) - I(V_0; S).$$

A pair (R, L) is achievable at distortion D if and only if $(R + \epsilon, L + \epsilon) \in \mathcal{R}_C(D + \epsilon)$ for every $\epsilon > 0$.

The proof is in Appendix A-C. The converse single-letterizes directly for the continuous source, and the achievability reduces to the finite-alphabet theorem of Villard and Piantanida [1, Theorem 3] by quantization, along the lines of Appendix A. Since $V_0 - V - T - (C, S)$, the leakage bound can also be written $I(T; V) - I(V; C | V_0) - I(V_0; S)$. With C constant the best choice is $V_0 = V$, because $I(V_0; S) \leq I(V; S)$, and the bound becomes $I(T; V | S)$, the region of Theorem 3. In general the two auxiliaries are what makes the region hard to evaluate. For a matrix distortion on the whole source and a full-rank holder, Xu, Chen, Chen, and Jin [5] show that Gaussian auxiliaries are optimal. Under a weighted distortion we do not settle this. The two degraded cases are explicit.

For Gaussian linear observations the information matrices order the side informations. If $J_S \preceq J_C$, the sufficient statistic $J_C T + \nu_C$, $\nu_C \sim \mathcal{N}(0, J_C)$, of C is mapped by $A = J_S J_C^+$ to $J_S T + A \nu_C$. Here $A J_C = J_S$, because $J_S \preceq J_C$ puts the null space of J_C inside that of J_S . Also $A J_C A^T = J_S^{1/2} (J_S^{1/2} J_C^+ J_S^{1/2}) J_S^{1/2} \preceq J_S$, because $J_C^{+1/2} J_S J_C^{+1/2} \preceq J_C^{+1/2} J_C J_C^{+1/2} \preceq I$ gives $\|X\| \leq 1$ for $X = J_S^{1/2} J_C^{+1/2}$, so that $XX^T = J_S^{1/2} J_C^+ J_S^{1/2} \preceq I$. So adding independent noise of covariance $J_S - A J_C A^T$ reproduces the law, given T , of the sufficient statistic $J_S T + \nu_S$ of S , and drawing S' from the conditional law of S given that statistic, which does not depend on T , gives a channel from C to S' with $(T, S') \sim (T, S)$ and $T - C - S'$. For any V_0 with $V_0 - T - (C, S)$, the mutual information $I(V_0; S)$ depends only on the law of (V_0, T) and the channel from T to S , so $I(V_0; S) = I(V_0; S') \leq I(V_0; C)$ by data processing. Applied to the conditional laws given V_0 , the same argument gives $I(V; S | V_0) \leq I(V; C | V_0)$ when $V_0 - V - T - (C, S)$. The symmetric statements hold when $J_C \preceq J_S$.

Lemma 30. *Let $V - T - (C, S)$ with $J_C \preceq J_S$, and let $M_C = \mathbb{E}[\text{Cov}(T | V, C)]$ and $M_S = \mathbb{E}[\text{Cov}(T | V, S)]$ be the minimum mean-square error covariances, with $M_C \succ 0$. Then $M_S \preceq (M_C^{-1} + J_S - J_C)^{-1}$, with equality when (V, T) is jointly Gaussian.*

Proof. Write $J_+ = J_S - J_C \succeq 0$. The conditional law of T given (V, S) depends on S only through its sufficient statistic $J_S T + \nu_S$, $\nu_S \sim \mathcal{N}(0, J_S)$, which has the law, given T , of the sum $c + o$ of the sufficient statistics $c = J_C T + \nu_C$ and $o = J_+ T + \nu_+$ of two independent observations C' and O of T with information matrices J_C and J_+ . Take $C' = H_C T + N'_C$ with $N'_C \sim \mathcal{N}(0, \Sigma_{N_C})$ independent of (V, T, U) and of the noise of O below. Since $c + o$ is sufficient for T in (C', O) and $V - T - (C', O)$, the error covariance M_S equals that of estimating T from (V, C', O) , and (T, V, C') has the law of (T, V, C) . Write $J_+ = B^T B$ and take $O = B T + n$ with $n \sim \mathcal{N}(0, I)$ independent of everything else, which has sufficient statistic $B^T O$ of the required law. Let $w = (V, C')$, let C_w be the conditional covariance of T given w , so that $\mathbb{E}[C_w] = M_C$, and fix the gain $L = M_C B^T (B M_C B^T + I)^{-1}$. The affine estimate $\mathbb{E}[T | w] + L(O - B \mathbb{E}[T | w])$ has conditional error covariance $(I - LB) C_w (I - LB)^T + LL^T$ given w , which is affine in C_w . Its mean is therefore $(I - LB) M_C (I - LB)^T + LL^T = M_C - M_C B^T (B M_C B^T + I)^{-1} B M_C = (M_C^{-1} + J_+)^{-1}$, and since M_S is the minimum mean-square error from (w, O) , $M_S \preceq (M_C^{-1} + J_+)^{-1}$. When (V, T)

is jointly Gaussian, $C_w = M_C$ for every w and the conditional mean of T given (w, O) is this affine estimate, so equality holds. $\square$

Theorem 31. *Let $0 < D < \text{tr}(Q\Sigma_{T|C})$.*

- (a) *If $J_S \preceq J_C$, the achievable pairs at distortion D are the quadrant above $(R_C(D), R_C(D))$, where $R_C(D) = \min\{\frac{1}{2} \log(\det \Sigma_{T|C} / \det M) : 0 \prec M \preceq \Sigma_{T|C}, \text{tr}(QM) \leq D\}$ is reverse water-filling on the eigenvalues of $\Sigma_{T|C}^{1/2} Q \Sigma_{T|C}^{1/2}$. The leakage-optimal and rate-optimal descriptions coincide, and the holder learns as much as the rate.*
- (b) *If $J_C \preceq J_S$, the achievable pairs at distortion D are the region of Proposition 2 for the source $\mathcal{N}(0, \Sigma_{T|C})$ with holder information $J_S - J_C$ and the same weight Q . Its coordinate is the decoder's error covariance $M = \mathbb{E}[\text{Cov}(T | V, C)]$. The results of Sections III–VII are statements about that program, so they hold with Σ_T replaced by $\Sigma_{T|C}$, J by $J_S - J_C$ (written as $H'^T H'$ with unit noise where an observation map is needed), and $D_{\max}$ by $\text{tr}(Q\Sigma_{T|C})$.*

Proof. Let V be any description with $I(T; V) < \infty$ and write $M_C = \mathbb{E}[\text{Cov}(T | V, C)]$, which is positive definite because a singular M_C reveals a linear functional of T without noise and makes $I(T; V | C)$ infinite. Since T has a conditional density given C and $I(T; V | C) < \infty$, it has a conditional density given (V, C) almost surely. Since $h(T | C) = \frac{1}{2} \log \det(2\pi e \Sigma_{T|C})$ and, by the Gaussian maximum-entropy bound and the concavity of $\log \det$, $h(T | V, C) \leq \frac{1}{2} \log \det(2\pi e M_C)$, we have $I(T; V | C) \geq \frac{1}{2} \log(\det \Sigma_{T|C} / \det M_C)$, and likewise $I(T; V | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det M_S)$. The distortion of any reconstruction from (V, C) is at least $\text{tr}(QM_C)$.

(a) By Theorem 29, an achievable (R, L) has $(R + \epsilon, L + \epsilon) \in \mathcal{R}_C(D + \epsilon)$ for every $\epsilon > 0$. For any feasible (V_0, V) , $I(V_0; C) \geq I(V_0; S)$ because $J_S \preceq J_C$, so both $R + \epsilon$ and $L + \epsilon$ are at least $I(T; V | C) \geq R_C(D + \epsilon)$. The function R_C is continuous on $(0, \text{tr}(Q\Sigma_{T|C}))$, so $R, L \geq R_C(D)$. Conversely, V_0 constant and a jointly Gaussian V with $\text{Cov}(T | V, C)$ equal to the minimizer M of $R_C(D)$, with $\hat{y} = \mathbb{E}[Y | V, C]$, put $(R_C(D), R_C(D))$ in $\mathcal{R}_C(D)$, and the region is increasing in D . The minimizer exists because the objective grows without bound as $\det M \rightarrow 0$. Let $\theta \in (0, 1]$ make $\theta \Sigma_{T|C}$ feasible, and take $\delta > 0$ so small that the objective exceeds its value at $\theta \Sigma_{T|C}$ wherever $\det M < \delta$. The minimum may then be taken over the compact feasible set with $\det M \geq \delta$, which is nonempty and on which the objective is continuous. Such a V exists because $M^{-1} - \Sigma_{T|C}^{-1} \succeq 0$.

(b) For feasible (V_0, V) , $J_C \preceq J_S$ gives $I(V; S | V_0) \geq I(V; C | V_0)$, so $I(V_0; C) - I(V_0; S) \geq I(V; C) - I(V; S)$ and the leakage bound is at least $I(T; V | C) + I(V; C) - I(V; S) = I(T; V | S)$, with equality for $V_0 = V$. By the bounds above and Lemma 30, $R + \epsilon \geq \frac{1}{2} \log(\det \Sigma_{T|C} / \det M_C)$, $L + \epsilon \geq \frac{1}{2} \log \det \Sigma_{T|S} + \frac{1}{2} \log \det(M_C^{-1} + J_S - J_C)$, and $\text{tr}(QM_C) \leq D + \epsilon$, with $0 \prec M_C \preceq \Sigma_{T|C}$. Since $(\Sigma_{T|C}^{-1} + J_S - J_C)^{-1} = \Sigma_{T|S}$, these are the coordinates (1) of the source $\mathcal{N}(0, \Sigma_{T|C})$ with holder information $J_S - J_C$ at $\Sigma_{e0} = M_C$, at distortion $D + \epsilon$. The compactness argument in the proof of Theorem 3 removes ϵ . Conversely, $V_0 = V$ jointly Gaussian with $\text{Cov}(T | V, C) = M$ for any $0 \prec M \preceq \Sigma_{T|C}$ attains all three coordinates with equality, by the equality case of Lemma 30, and Theorem 29 makes them achievable.

The substitution in (b) is exact for three reasons. First, the program of Proposition 2 depends on the holder only through J , and so does every expression in Sections III–VII that uses H and Σ_U , because $\det(HXH^T + \Sigma_U) = \det \Sigma_U \det(I + JX)$ for every X . Any factorization $J_S - J_C = H'^T H'$ with unit noise therefore gives the same program. Second, a jointly Gaussian description $V = a^T T + N$ with $N \sim \mathcal{N}(0, \sigma^2)$ independent of (T, C, S) has $\text{Cov}(T | V, C) = (\Sigma_{T|C}^{-1} + aa^T / \sigma^2)^{-1}$, the error covariance it would have for the source $\mathcal{N}(0, \Sigma_{T|C})$. So statements about linear Gaussian descriptions carry over with $\mathbb{E}[Y | \hat{Y}]$ replaced by $\mathbb{E}[Y | V, C]$. Third, every constant and hypothesis of those sections is read on the substituted parameters. In particular Σ_Y becomes $F\Sigma_{T|C}F^T$, the decoder's remaining uncertainty about Y , and a hypothesis such as a holder that observes only the context becomes a condition on $J_S - J_C$. $\square$

So a decoder that knows more than the holder removes the tradeoff, and a holder that knows more than the decoder leaves exactly the problem of the preceding sections for the source as the decoder already knows it, with the holder's excess information in place of its total. When neither dominates, the auxiliary V_0 can matter, and the region is beyond the methods of this paper.

X. INSTANCES

Scalar. At $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$ the rate-optimal channel pays 0.0349 bits of excess leakage and the leakage-optimal channel 0.0400 bits of excess rate. Over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$ the rate excess of the leakage-optimal channel stays below 0.114 bits and the leakage excess of the rate-optimal channel below 0.077 bits, and at $(\rho^2, \tau^2, D) = (0.5, 10^{-3}, 0.5)$ the rate excess reaches 1.537 bits. Two sources with the same law of (Y, S) up to scaling of S , $(\rho^2, \tau^2) = (\frac{3}{5}, \frac{1}{5})$ and $(\frac{3}{4}, \frac{1}{2})$, have $L = 1.1880$ and 1.2105 bits at $D = 0.1$. The leakage is not determined by the law of (Y, S) , because the encoder observes the context.

Coupled. For the coupled instance of Section II, the certificate of Theorem 16(b) holds for $D \geq 1.08530$, out of $D_{\max} = 1.4$. The rate-optimal description reduces the error along the first axis for every $D \geq 0.8$. The leakage-optimal description reduces it along 47.48° , 46.76° , 45.07° , and 41.15° at $D = 1.38, 1.3, 1.2$, and 1.1 . Four sources with the same u and the same law of (Y, S) up to scaling, $(|\text{Cov}(Y, V)|^2, \tau^2) = (0.33, 0.1)$, $(0.36, 0.2)$, $(0.39, 0.3)$, $(0.42, 0.4)$, read along 42.83° , 37.61° , 33.75° , and 30.84° at $D = 1.3$, each inside the range of its certificate. The direction, too, is not determined by the law of (Y, S) .

Independent components. With $(\sigma_1^2, \rho_1^2, \tau_1^2) = (1, 0, 1)$ and $(\sigma_2^2, \rho_2^2, \tau_2^2) = (0.6, 0.9, 0.05)$, $\kappa_1 = 0.7213$, $\kappa_2 = 0.0633$, and the two descriptions describe disjoint components for every $D \in [1.2002, 1.6)$. At $D = 1.4$ the rate-optimal description pays $R = L = 0.1610$ bits on component 1 and the leakage-optimal description pays $L = 0.0197$ and $R = 0.3272$ bits on component 2.

XI. DISCUSSION

A description held by two parties has a cost for each, and the encoder's observations decide how the costs trade off. For Gaussian sources without decoder side information the region is now explicit in structure. It is a convex program in the error covariance, it has a closed form in a one-dimensional case, and its directions turn with the budget exactly when the two parties' geometries are not aligned. Several problems remain open. With side information at the decoder, the region is proved for the continuous source and the degraded cases are settled here. When neither side information dominates, the second auxiliary can matter. For a matrix distortion on the whole source and a full-rank eavesdropper, Xu, Chen, Chen, and Jin [5] show that Gaussian auxiliaries are optimal. Whether their two-precision program, under a weighted distortion and a rank-deficient holder, has the convexity, pencil, and turning structure of the case without decoder side information is open. Explicit solutions are now available in four settings. They give the whole frontier in commuting geometry at every budget, the minimum leakage in arbitrary geometry in the one-dimensional case at large budgets, the whole frontier in arbitrary geometry at small budgets when the decoder reproduces the whole source, and the whole frontier for a scalar described variable with any context and any holder, up to one scalar root. In every geometry and at every budget the frontier is the fixed point of reverse water-filling on a tilted weight. For a scalar holder this is two scalar equations, and for a holder of any rank the fixed point is the limit of a monotone iteration of water-filling steps. A closed form for a holder of higher rank and a described variable of higher dimension, or for the boundaries between these cases, is not known. The general case with side information at both receivers, where the second auxiliary of Villard and Piantanida can matter, under a trace-type distortion and a rank-deficient holder, is where the main open question lies. Stationary sources and finite blocklengths are not treated.

APPENDIX A THE CODING THEOREM

A. Finite alphabets

Let (X, S) be a discrete memoryless pair with finite alphabets and let d be a distortion with values in $[0, d_{\max}]$. Codes, rate, distortion, and leakage $H(M | S^n)/n$ are as in Section II.

Theorem 32 ([1, Theorem 3], without decoder side information). *The closure of the achievable set at distortion D is $\bigcup_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}$, and the union is convex and closed.*

Proof outline. Villard and Piantanida prove the region in the equivocation form $\Upsilon \leq H(X | V) + 0 - I(X; S | U)$ for the equivocation Υ , $U - V - X - S$, with no decoder side information. Taking $U = V$ is optimal, because $I(X; S | U) = I(V; S | U) + I(X; S | V) \geq I(X; S | V)$, and $\Upsilon = H(X | S) - L$ for a deterministic encoder, which gives the displayed form with V in place of $\hat{X}$, and a reproduction that is a function of V only lowers both

coordinates. For the Gaussian extension we use the following finite-blocklength form of the achievability, obtained by covering at rate $I(X; \hat{X}) + 2\epsilon$, a random binning of the codeword index into $2^{n(I(X; \hat{X}|S) + 3\epsilon)}$ bins used only as a proof device, Fano's inequality for a hypothetical decoder holding the bin index and S^n , and selection of one deterministic code. For every channel with finite reproduction alphabet and every $\epsilon > 0$ there are $\epsilon_n \rightarrow 0$ and, for all large n , a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X; \hat{X}) + 2\epsilon)}$, distortion at most $\mathbb{E}d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\max}$, and $H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$. Convexity of both coordinates in the channel, for the conditional one by the decomposition $I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q)$, makes the union convex without time sharing. $\square$

B. The Gaussian source

Proof of Theorem 3. Converse. For any code, $\log |\mathcal{M}_n| \geq I(T^n; \hat{\mathbf{Y}}^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i)$ and $H(M | S^n) \geq I(T^n; \hat{\mathbf{Y}}^n | S^n) = \sum_i [I(T_i; \hat{\mathbf{Y}}^n, T^{i-1}, S^n) - I(T_i; S_i)] \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i | S_i)$, by the chain rule and the independence of the pairs (T_i, S_i) . Every term is finite, being at most $I(T_i; S_i) + \log |\mathcal{M}_n|$ with $I(T_i; S_i) = \frac{1}{2} \log \det(H\Sigma_T H^\top + \Sigma_U) / \det \Sigma_U$. The mixture of the per-letter channels satisfies the Markov chain, has distortion D_n , and by convexity of both coordinates in the channel (for general alphabets the conditional coordinate is an integral over s of relative entropies between measures that depend affinely on the channel, and relative entropy is jointly convex in its two arguments [30], [29]) lies below (R_n, L_n) , so (R_n, L_n) is in the union at D_n .

Closedness. Let $(R_j, L_j) \rightarrow (R, L)$ lie in the union at levels D_j with $\limsup D_j \leq D$, with $\Sigma_j \in \mathcal{P}(D_j)$ below them by Proposition 2. The Σ_j lie in the compact set $0 \leq \Sigma \leq \Sigma_T$ and $\det \Sigma_j \geq \det \Sigma_T 2^{-2R_j}$ is bounded away from zero, so a subsequence converges to some $\Sigma^* \succ 0$ in $\mathcal{P}(D)$. Since r and ℓ are continuous on positive definite matrices, (R, L) lies above $(r(\Sigma^*), \ell(\Sigma^*))$.

Achievability. Fix an admissible channel with $I(T; \hat{\mathbf{Y}}) < \infty$. Then $\mathbb{E}\|\hat{\mathbf{Y}}\|^2 < \infty$ because its distortion is finite and $W \succ 0$. Let q_k be the saturating dyadic quantizer of level k applied coordinatewise. Its bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, map to their left endpoints and the two tail cells to $\pm 2^k$. The partitions are nested and generate the Borel σ -field.

Step A. The σ -fields of $S_\eta := q_\eta(S)$ increase to $\sigma(S)$, so $I(\hat{\mathbf{Y}}; S_\eta) \uparrow I(\hat{\mathbf{Y}}; S)$ [27, Ch. 2]. Fix η with $I(\hat{\mathbf{Y}}; S_\eta) \geq I(\hat{\mathbf{Y}}; S) - \epsilon$. Since $\hat{\mathbf{Y}} - T - S_\eta$, $I(T; \hat{\mathbf{Y}} | S_\eta) = I(T; \hat{\mathbf{Y}}) - I(\hat{\mathbf{Y}}; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + \epsilon$.

Step B. With $T_k = q_k(T)$ and $\hat{\mathbf{Y}}_k = q_k(\hat{\mathbf{Y}})$, the surrogate source (T_k, S_η) is discrete memoryless, the surrogate channel is $p(\hat{\mathbf{y}} | t_k) = \Pr(\hat{\mathbf{Y}}_k = \hat{\mathbf{y}} | T_k = t_k)$, and the surrogate distortion $d_k(t_k, \hat{\mathbf{y}}) = \mathbb{E}[(Y - \hat{\mathbf{y}})^\top W (Y - \hat{\mathbf{y}}) | T_k = t_k]$ is bounded for each k , every cell having positive probability since $\Sigma_T \succ 0$. The finite-blocklength achievability of Theorem 32 gives a code of rate at most $I(T_k; \hat{\mathbf{Y}}_k) + 2\epsilon \leq I(T; \hat{\mathbf{Y}}) + 2\epsilon$. Its true distortion equals its surrogate distortion, because given T_k^n the law of Y_i depends on $T_{k,i}$ alone. The surrogate distortion of the channel converges to $\mathbb{E}[(Y - \hat{\mathbf{Y}})^\top W (Y - \hat{\mathbf{Y}})] \leq D$, because $\mathbb{E}[Y | T_k] \rightarrow Y$ in L^2 by martingale convergence and $\hat{\mathbf{Y}}_k \rightarrow \hat{\mathbf{Y}}$ in L^2 . The surrogate law of $(\hat{\mathbf{Y}}_k, S_\eta)$ converges weakly to the true law of $(\hat{\mathbf{Y}}, S_\eta)$, by the bounded-convergence and martingale argument applied to test functions bounded on $\mathbb{R}^m \times q_\eta(\mathbb{R}^r)$ and continuous in the first argument, and the Markov chain $\hat{\mathbf{Y}} - T - S$. Mutual information is lower semicontinuous under weak convergence [28], so $I_k(T_k; \hat{\mathbf{Y}}_k | S_\eta) = I(T_k; \hat{\mathbf{Y}}_k) - I_k(\hat{\mathbf{Y}}_k; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + 2\epsilon$ for large k , and the code has $H(M | S_\eta^n) \leq n(I(T; \hat{\mathbf{Y}} | S) + 5\epsilon + o(1))$.

Step C. S_η^n is a function of S^n , so $H(M | S^n) \leq H(M | S_\eta^n)$. A diagonal sequence $\epsilon \downarrow 0$ achieves the corner $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ at distortion D , and with the converse, closedness, and Proposition 2 the region is as stated. $\square$

C. Side information at the decoder

Finite alphabets. Let (X, C, E) be a discrete memoryless source with finite alphabets, the encoder seeing X^n , the decoder (M, C^n) , and the holder (M, E^n) , and let the distortion take values in $[0, d_{\max}]$. Villard and Piantanida [1, Theorem 3] prove, for deterministic encoders, that the closure of the achievable rate–distortion–equivocation triples is the closure of the set of (R, D, Υ) with $R \geq I(X; V | C)$, $D \geq \mathbb{E}d(X, \hat{x}(V, C))$, and $\Upsilon \leq H(X | V, C) + I(X; C | V_0) - I(X; E | V_0)$ for finite auxiliaries with $V_0 - V - X - (C, E)$, where the equivocation is $\frac{1}{n}H(X^n | M, E^n)$. Their reproduction alphabet is the source alphabet. A finite reproduction alphabet $\mathcal{Y}$ is covered by enlarging the source alphabet to the disjoint union $\mathcal{X} \sqcup \mathcal{Y}$, with the added letters of probability zero. For $x \in \mathcal{X}$ and $b \in \mathcal{X}$ set $d(x, b) = \max_{x', y} d(x', y)$, and let d be bounded but otherwise arbitrary on the added source letters,

so that replacing any such output by a fixed $y_0 \in \mathcal{Y}$ does not increase the distortion. For a deterministic encoder the leakage is $\frac{1}{n}H(M | E^n) = H(X | E) - \frac{1}{n}H(X^n | M, E^n)$, and since $V_0 - X - (C, E)$,

$$H(X | E) - H(X | V, C) - I(X; C | V_0) + I(X; E | V_0) = I(X; V | C) + I(V_0; C) - I(V_0; E).$$

So for every point $(I(X; V | C), \mathbb{E}d(X, \hat{x}(V, C)), I(X; V | C) + I(V_0; C) - I(V_0; E))$ of finite auxiliaries and every $\epsilon > 0$, some deterministic code has rate, distortion, and leakage each within ϵ of that point or below it.

Proof of Theorem 29. Converse. Fix a code, so $M = f_n(T^n)$ and $\hat{Y}^n = g_n(M, C^n)$. For $i = 1, \dots, n$ let

$$V_{0,i} = (M, C^{i-1}, S_{i+1}^n), \quad V_i = (M, C^{i-1}, C_{i+1}^n, S_{i+1}^n).$$

Then $V_{0,i}$ is a function of V_i , and $V_i - T_i - (C_i, S_i)$ because, given T_i , the variable V_i is a function of T_i and of (T^n, C^n, S^n) with the i -th letter removed, which is independent of (T_i, C_i, S_i) . The reconstruction $\hat{Y}_i$ is a function of (V_i, C_i) . Every mutual information below is finite, being at most $H(M)$ or $I(T_i; C_i)$ or $I(T_i; S_i)$, so the chain rule and the Csiszár sum identity apply to these continuous variables. For the rate,

$$nR_n \geq H(M) \geq I(T^n; M | C^n) = \sum_i I(T_i; M, C_{\setminus i}^n, T_{i+1}^n | C_i) \geq \sum_i I(T_i; V_i | C_i),$$

where the equality uses the chain rule in reverse order and the independence of T_i from $(C_{\setminus i}^n, T_{i+1}^n)$ given C_i , and the inequality uses data processing, since S_{i+1}^n is obtained from T_{i+1}^n through noise independent of everything else. For the leakage, $H(M | T^n, S^n) = 0$ and $M - T^n - (C^n, S^n)$ give

$$nL_n = I(M; T^n | S^n) = I(M; T^n | C^n) + I(M; C^n) - I(M; S^n),$$

and the Csiszár sum identity gives $I(M; C^n) - I(M; S^n) = \sum_i [I(M; C_i | C^{i-1}, S_{i+1}^n) - I(M; S_i | C^{i-1}, S_{i+1}^n)]$. Since (C^{i-1}, S_{i+1}^n) is independent of (C_i, S_i) , each term equals $I(V_{0,i}; C_i) - I(V_{0,i}; S_i)$, so $nL_n \geq \sum_i [I(T_i; V_i | C_i) + I(V_{0,i}; C_i) - I(V_{0,i}; S_i)]$. With τ uniform on $\{1, \dots, n\}$ and independent of everything, set $V_0 = (V_{0,\tau}, \tau)$, $V = (V_\tau, \tau)$, and $(T, C, S) = (T_\tau, C_\tau, S_\tau)$, which has the source law. Then $V_0 - V - T - (C, S)$, $I(T; V) \leq H(M) < \infty$, $I(T; V | C)$, $I(V_0; C)$, and $I(V_0; S)$ are the averages of their per-letter versions, and the reconstruction $\hat{y}(V, C) = \hat{Y}_\tau$ has distortion D_n . So $(R_n, L_n) \in \mathcal{R}_C(D_n)$ for every code. If (R, L) is achievable at D , then for every $\epsilon > 0$ and all large n some code has $R_n \leq R + \epsilon$, $L_n \leq L + \epsilon$, $D_n \leq D + \epsilon$, and since $\mathcal{R}_C$ is increasing in (R, L) and in D , $(R + \epsilon, L + \epsilon) \in \mathcal{R}_C(D + \epsilon)$.

Achievability. It suffices to show that for every feasible $(V_0, V, \hat{y})$ and $\epsilon > 0$ there are codes, for all large n , with rate at most $I(T; V | C) + \epsilon$, distortion at most $D + \epsilon$, and leakage at most $I(T; V | C) + I(V_0; C) - I(V_0; S) + \epsilon$. A diagonal sequence over ϵ then achieves any (R, L) with $(R + \epsilon, L + \epsilon) \in \mathcal{R}_C(D + \epsilon)$ for all ϵ . Replace $\hat{y}$ by $\mathbb{E}[Y | V, C]$, which does not increase the distortion. The auxiliaries V_0 and V take values in Euclidean spaces, as in the definition of $\mathcal{R}_C(D)$, and the auxiliaries built in the converse do too, because M and τ take finitely many values and, after zero-padding V_i and $V_{0,i}$ to a common dimension, which stays injective once τ is included, each lies in a fixed Euclidean space. Let q_j denote the saturating dyadic quantizers of Appendix A.

Step A (auxiliaries and side informations). Let $U_j = q_j(V_0)$, $V'_j = (U_j, q_j(V))$, $C_\kappa = q_\kappa(C)$, and $S_\eta = q_\eta(S)$. Then U_j is a function of V'_j and $U_j - V'_j - T - (C_\kappa, S_\eta)$. Along the nested partitions, $I(T; V'_j) \uparrow I(T; V_0, V) = I(T; V)$, $I(V'_j; C_\kappa) \uparrow I(V; C)$, $I(U_j; C_\kappa) \uparrow I(V_0; C)$, and $I(U_j; S_\eta) \uparrow I(V_0; S)$ [27, Ch. 2], all limits finite. Since $V - T - C$, $I(T; V | C) = I(T; V) - I(V; C)$, so $I(T; V'_j | C_\kappa) \rightarrow I(T; V | C)$ and the leakage expression with $(U_j, V'_j, C_\kappa, S_\eta)$ converges to the one with (V_0, V, C, S) . With $\hat{y}_{j,\kappa} = \mathbb{E}[Y | V'_j, C_\kappa]$, martingale convergence gives $\hat{y}_{j,\kappa} \rightarrow \mathbb{E}[Y | V_0, V, C] = \mathbb{E}[Y | V, C]$ in L^2 , because $V_0 - (V, C) - T$, so the distortion converges as well. Fix j, κ, η with all three quantities within ϵ of their limits. The values of $\hat{y}_{j,\kappa}$ form a finite set $\mathcal{Y}$ with $\max_{y \in \mathcal{Y}} \|y\|_W =: b < \infty$.

Step B (the surrogate source). Let $T_k = q_k(T)$ and let the surrogate discrete memoryless source be $(X, C, E) = (T_k, C_\kappa, S_\eta)$ with its true joint law. A code that uses only T_k^n at the encoder and (M, C_κ^n) at the decoder is a code for the Gaussian problem whose rate and leakage $H(M | S_\eta^n)$ equal its surrogate rate and leakage, because the pair (T_k^n, S_η^n) has the same law in both. Give the surrogate the distortion $d_k(t, y) = \mathbb{E}[(Y - y)^T W (Y - y) | T_k = t]$ and the auxiliaries the channel $p_k(u, v' | t) = \Pr(U_j = u, V'_j = v' | T_k = t)$, which keeps U_j a function of V'_j and so the chain $U - V' - X - (C, E)$. The true distortion of a code differs from its surrogate distortion only because the decoder's output depends on C_κ^n . With $m(t) = \mathbb{E}[Y | T_k = t]$ and $m(t, c) = \mathbb{E}[Y | T_k = t, C_\kappa = c]$, the conditional expected distortion of an output y given $(T_{k,i}, C_{k,i}) = (t, c)$ is $\|y - m(t, c)\|_W^2 + v(t, c)$ against $\|y - m(t)\|_W^2 + v(t)$

for d_k , and averaging over c given t cancels every term except $-2\mathbb{E}[y_i^\top W(m(T_k, C_\kappa) - m(T_k))]$. By Cauchy–Schwarz the difference per letter is at most $2b\mathbb{E}[\|m(T_k, C_\kappa) - m(T_k)\|_W^2]^{1/2}$, which tends to 0 as $k \rightarrow \infty$ because both conditional means converge to Y in L^2 , and the bound holds for every code whose outputs lie in $\mathcal{Y}$.

Step C (convergence of the surrogate quantities). The surrogate joint law of $(U_j, V'_j, C_\kappa, S_\eta)$ is $\mathbb{E}[\mathbf{1}\{C_\kappa = c, S_\eta = s\} \Pr(U_j = u, V'_j = v' \mid T_k)]$, which converges as $k \rightarrow \infty$ to $\mathbb{E}[\mathbf{1}\{C_\kappa = c, S_\eta = s\} \Pr(U_j = u, V'_j = v' \mid T)]$, the true joint law, by martingale convergence and $(V_0, V) - T - (C, S)$. All four alphabets are finite, so the surrogate values of $I(V'; C)$, $I(U; C)$, and $I(U; E)$ converge to the true ones, while the surrogate $I(X; V') = I(T_k; V'_j) \leq I(T; V'_j)$. The surrogate distortion of $\hat{x}(v', c) = \hat{y}_{j, \kappa}(v', c)$ is $\mathbb{E}[\sum_{v'} \Pr(V'_j = v' \mid T_k) (\|\hat{y}_{j, \kappa}(v', C_\kappa) - m(T_k)\|_W^2 + v(T_k))]$, which converges to $\mathbb{E}\|Y - \hat{y}_{j, \kappa}(V'_j, C_\kappa)\|_W^2$ because $m(T_k) \rightarrow Y$ in L^2 , $\mathbb{E}v(T_k) \rightarrow 0$, and the outputs are bounded. Fix k with the surrogate rate, leakage, and distortion within ϵ of their values in Step A and the distortion mismatch of Step B at most ϵ .

Step D (codes). By the finite-alphabet theorem there is a blocklength n_0 and a deterministic code of that length with rate at most $I(T; V \mid C) + 4\epsilon$, surrogate distortion at most $D + 4\epsilon$, outputs in $\mathcal{Y}$, and $\frac{1}{n_0}H(M \mid S_\eta^{n_0})$ at most the leakage expression plus 4ϵ . As a code for the Gaussian problem its distortion is at most $D + 5\epsilon$ by Step B, and its leakage satisfies $H(M \mid S_\eta^{n_0}) \leq H(M \mid S_\eta^{n_0})$ because $S_\eta^{n_0}$ is a function of S^{n_0} . For a general blocklength $n = kn_0 + r$ with $0 \leq r < n_0$, apply this code to each of the k blocks and reproduce the last r letters by zero. The message is the tuple of block messages, so the rate does not increase. The pairs of block message and block of S^n are independent across blocks, so the leakage is the sum of the block leakages divided by n and does not exceed that of one block. The distortion is at most $(kn_0(D + 5\epsilon) + rD_{\max})/n$, whose limsup is at most $D + 5\epsilon$. $\square$

APPENDIX B THE GAUSSIAN EXHAUSTION LEMMA

Proof of Lemma 1. The divergence bound. Conditional mutual information is the expected divergence of the conditional law of T given $(\hat{\mathbf{Y}}, S)$ from the conditional law given S [29], [27]. Fix a realization, let P be the conditional law with mean μ and covariance C , $P_G = \mathcal{N}(\mu, C)$, and $Q_S = \mathcal{N}(\mu_S, \Sigma_{T|S})$. If P has no density the divergence is infinite. Otherwise $\log(dP_G/dQ_S)$ is quadratic in t and P, P_G share two moments, so $D(P\|Q_S) = D(P\|P_G) + D(P_G\|Q_S) \geq D(P_G\|Q_S)$. Averaging $D(P_G\|Q_S)$, the trace term plus the mean term, $\mathbb{E}[\text{tr}(\Sigma_{T|S}^{-1}C) + (\mu - \mu_S)^\top \Sigma_{T|S}^{-1}(\mu - \mu_S)]$, averages to p by the law of total covariance, and by concavity of $\ln \det$ and Jensen, $\mathbb{E}[-\ln \det C] \geq -\ln \det \mathbb{E}C \geq -\ln \det \Sigma_e$, since the conditional mean has error covariance no larger than any linear estimator. Hence $I(T; \hat{\mathbf{Y}} \mid S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the same argument without S gives $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$.

The determinant identity. With $\hat{\mathbf{Y}} = AT + N'$, $\text{Cov}(N', T) = 0$, and $\text{Cov}(N', S) = 0$, the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block triangular with unit diagonal, so $\det \text{Cov}(T, S, \hat{\mathbf{Y}}) = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N$. The Schur factorization in the other order gives $\det \text{Cov}(\hat{\mathbf{Y}}, S) \det \Sigma_e$, and $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S) \Sigma_S^{-1} \text{Cov}(S, \hat{\mathbf{Y}}) = A \Sigma_{T|S} A^\top + \Sigma_N \succ 0$. So $\det \Sigma_e = \det \Sigma_{T|S} \det \Sigma_N / \det(A \Sigma_{T|S} A^\top + \Sigma_N)$, finite and positive.

Equality. For the jointly Gaussian channel $AT + N$ the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with the linear estimate as mean and Σ_e as covariance, so every inequality is an equality. Its second moments match the given channel's. $\square$

APPENDIX C PRACTICAL QUANTIZERS

The theorems concern long blocks and Gaussian test channels. A practical description is often a fixed-rate scalar quantizer applied to one projection of the data. We ask two questions of such a quantizer. Does the projection it reads change its leakage at equal rate and under the same distortion budget? And does the leakage-aware projection turn with the budget, as Theorem 20 predicts for the optimum?

Setup. Take the coupled instance of Section II, so that $Q = F^\top F$ and $D_{\max} = \text{tr} \Sigma_Y = 1.4$. For a direction b let $Z = b^\top T / (b^\top \Sigma_T b)^{1/2}$, which is standard normal, and let $M = q(Z)$ for a scalar quantizer q with K levels and thresholds at the midpoints between levels. The decoder reproduces $\hat{\mathbf{Y}} = \mathbb{E}[Y \mid M]$. Write p_i for the probability of cell i , m_i for $\mathbb{E}[Z \mid M = i]$, and $\gamma = \sum_i p_i m_i^2$. Let $c = \mathbb{E}[YZ] = F^\top \Sigma_T b / (b^\top \Sigma_T b)^{1/2}$ and let ρ be the correlation of Z with the scalar S . Because $Y - cZ$ is independent of Z , the distortion is

$$D(b) = \text{tr} \Sigma_Y - \gamma \|c\|^2.$$

TABLE I

LLOYD-MAX QUANTIZERS ON ONE PROJECTION OF THE COUPLED INSTANCE. AT EACH BUDGET THE LEAKAGE-AWARE PROJECTION LEAKS LESS THAN THE PRINCIPAL AXIS AT THE SAME RATE, AND IT READS A DIRECTION THAT TURNS FURTHER THAN THE GAUSSIAN OPTIMUM. LEAKAGE IN BITS, ANGLES IN DEGREES IN THE PLANE OF Y .

k	$H(M)$	f	D_0	principal	aware	reduction	angle (aware / pencil)	$L_{1d}(D_0)$
2	1.9111	0.15	0.6499	1.8147	1.4205	21.7%	16.40 / 6.83	0.8758
		0.40	0.8705	1.8147	0.8956	50.6%	32.09 / 20.04	0.3913
		0.70	1.1352	1.8147	0.5066	72.1%	49.93 / 42.94	0.0636
3	2.8249	0.15	0.5794	2.7188	2.2667	16.6%	16.50 / 4.36	1.1201
		0.40	0.8207	2.7188	1.5896	41.5%	31.78 / 16.09	0.4803
		0.70	1.1104	2.7188	0.9714	64.3%	50.04 / 41.73	0.0792
4	3.7653	0.15	0.5581	3.6563	3.1855	12.9%	16.58 / 3.71	1.2126
		0.40	0.8057	3.6563	2.4463	33.1%	32.02 / 15.02	0.5088
		0.70	1.1029	3.6563	1.6851	53.9%	50.01 / 41.32	0.0845

The entropy $H(M) = -\sum_i p_i \log p_i$ does not depend on b , so every comparison below is at equal rate, whether the index is sent at $\log K$ bits or entropy coded. Given S , the variable Z is normal with variance $1 - \rho^2$ and a mean proportional to ρ , so the leakage $H(M | S)$ depends on b only through ρ .

Proposition 33. *For every quantizer q , the leakage $H(q(Z) | S)$ is a nonincreasing function of $|\rho|$.*

Proof. Let $\bar{S}$ be S scaled to unit variance. The law of $(Z, \bar{S})$ is fixed by ρ , and replacing $\bar{S}$ by $-\bar{S}$ flips the sign of ρ without changing $H(M | \bar{S})$, so the leakage is a function of $|\rho|$. For $0 \leq \rho < \rho' \leq 1$, let $(Z, \bar{S}')$ have correlation ρ' and let $\bar{S}'' = (\rho/\rho')\bar{S}' + (1 - \rho^2/\rho'^2)^{1/2}N$ with N standard normal and independent. Then $(Z, \bar{S}'')$ has correlation ρ and $Z - \bar{S}' - \bar{S}''$ is a Markov chain, so $H(M | \bar{S}'') \geq H(M | \bar{S}', \bar{S}'') = H(M | \bar{S}')$. $\square$

For every quantizer used here the leakage is strictly decreasing in $|\rho|$, which we checked at 200 values of $|\rho|$ in $[0, 0.995]$. The leakage-aware projection for a given quantizer therefore maximizes $|\rho|$ subject to $\gamma \|c\|^2 \geq \text{tr } \Sigma_Y - D_0$. We place each budget relative to the principal-axis quantizer, which reads Y_1 , has $c = (1, 0)$, and has distortion $D_R = \text{tr } \Sigma_Y - \gamma$. Setting $D_0 = D_R + f(\text{tr } \Sigma_Y - D_R)$ with $0 < f < 1$, the constraint becomes $\|c\|^2 \geq 1 - f$, and the quantizer has dropped out. At a given f the leakage-aware projection is the same for every quantizer, and only its leakage depends on the quantizer. For $f \geq 0.64$ it is the context itself. The choice $Z = V$ has $c = \text{Cov}(Y, V) = 0.6u$ with $\|c\|^2 = 0.36$, and V maximizes $|\rho|$ over all projections because $S = V + U$.

Computation. We use the Lloyd-Max quantizers for the standard normal law [31], [32] at $k = 2, 3$, and 4 bits, and $f = 0.15, 0.4$, and 0.7. The rate and the distortion are closed-form expressions in the cell probabilities and conditional means, which are evaluated through the normal distribution function. The leakage is an exact one-dimensional integral over S of the conditional entropy of the cell index, evaluated numerically by Gauss-Hermite quadrature with 80 nodes. The leakage-aware projection is the direction of largest $|\rho|$ among 2×10^6 random feasible directions, and its costs are then evaluated by the same formulas. Table I lists the results.

Results. At equal rate and equal budget, the leakage-aware quantizer leaks 12.9% to 72.1% less than the principal-axis quantizer (Table I and Fig. 1(a)). The gain grows as the budget loosens and is largest at 2 bits, where at $f = 0.7$ the quantizer leaks 0.5066 bits instead of 1.8147. The principal-axis quantizer reduces the error along the first axis at every budget. The leakage-aware quantizer reduces it along 16.4° to 16.6° , 31.8° to 32.1° , and 49.9° to 50.0° at the three values of f (Fig. 1(b)). The reduction above makes the direction the same for every quantizer, and the three agree within 0.31° , a spread left by the random search. They turn with the budget toward the context direction u at 50° , which the quantizer reaches once it reads V .

The Gaussian optimum of Theorem 16 turns the same way but less, through 3.7° to 6.8° , 15.0° to 20.0° , and 41.3° to 42.9° at the same budgets. A Gaussian description at budget D_0 chooses its noise variance as well as its direction, so it can spend part of the slack on coarser resolution. A fixed-rate quantizer has a fixed γ . It can spend the slack only on direction, and it turns further. Its leakage also stays well above the floor. Every budget at $f = 0.7$ lies in the certified range $D \geq 1.08530$ of Section X, where $L_{1d}(D_0) = L(D_0)$. Because $\hat{Y}$ is an admissible channel and $H(M | S) = I(T; M | S) \geq I(T; \hat{Y} | S)$, Theorem 3 makes $L(D_0)$ a lower bound on the leakage of the quantizer. The 2-bit quantizer leaks 0.5066 bits there, against a floor of 0.0636 bits. At $f = 0.15$

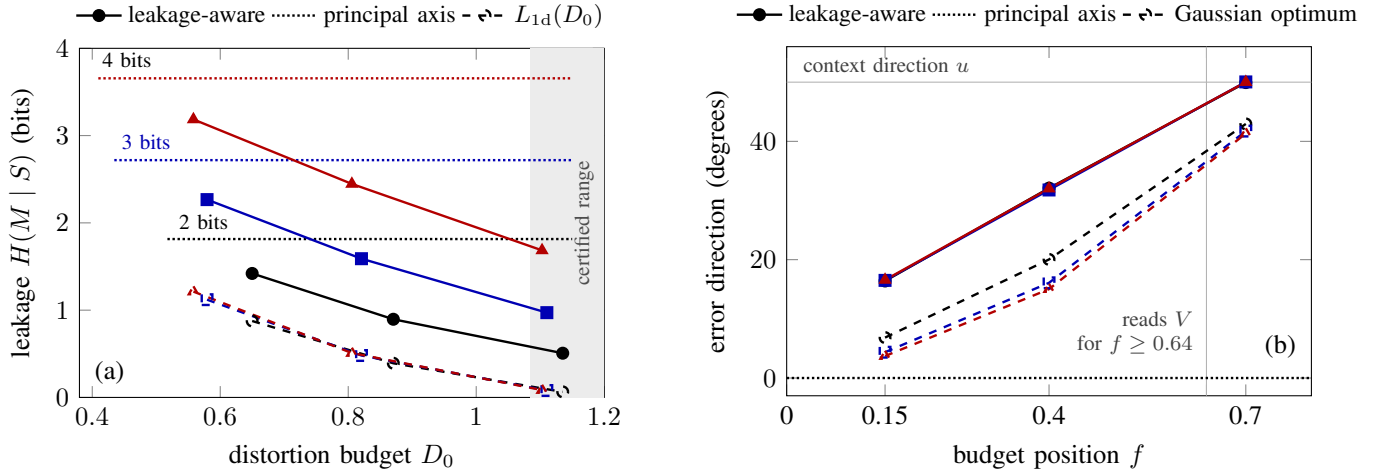


Fig. 1. Fixed-rate Lloyd–Max quantizers on one projection of the coupled instance, at 2 (circles), 3 (squares), and 4 bits (triangles). (a) Leakage at equal rate. The leakage-aware projection leaks less than the principal axis at every budget, and in the shaded certified range $L_{1d}(D_0)$ is the least leakage of any description. (b) The direction in the plane of Y along which each description reduces the error. The leakage-aware quantizer turns toward the context direction and turns further than the Gaussian optimum, while the principal-axis quantizer stays at 0° .

and 0.4 the budgets lie below the certified range, and $L_{1d}(D_0)$ is the best value among one-dimensional Gaussian descriptions rather than a bound.

APPENDIX D NUMERICAL CHECKS

The identities and instances were checked independently of the derivations by scripts run on a remote workstation. The scripts and their recorded outputs are in the public repository

<https://github.com/ahb-sjsu/geometric-observation>
commit 93d48f2ee53dff637bcb0590ba3fd6a065b9475c

in the directory `paper/tit-rate-leakage` unless another directory is named. `verify_general.py` (in the companion directory `tit-vector-described`) checks, for random sources with general weights and observation maps, that the pencil value of Theorem 16 equals the brute-force optimum over Gaussian descriptions whenever that optimum is one-dimensional, that both costs are convex in the error covariance (16000 midpoint tests), that the weighted minimizers are unique, the two directions of Theorem 20, and the criterion of Theorem 22. `verify_commuting.py` checks the closed form of Theorem 13 for the minimum leakage and for weighted frontier points against brute force, the diagonality of the optimum in the common eigenbasis, the activation rule, and the block-diagonal optimum of Theorem 12. `verify_coupled.py`, `verify_parallel.py` (in `tit-vector-described`) and `verify_pencil.py` (in `tit-cr-context`) check the scalar identity of Corollary 17 symbolically, the determinant program against brute force, the certificate, Lemma 26, the allocation, and the numbers of Section X. `tqp_leakage_demo.py` computes Table I and Fig. 1. It evaluates the rate and distortion of each quantizer in closed form and its leakage by 80-node Gauss–Hermite quadrature of an exact one-dimensional integral, and checks that the leakage is strictly decreasing in $|\rho|$ for every quantizer. Its output is recorded in `tqp_leakage_demo-2026-10-01.txt`. `verify_decoder_si.py` checks Lemma 30 for a non-Gaussian description (a sign quantizer, by numerical integration), the reduction of Theorem 31(b) on random Gaussian descriptions, and the minimum leakage with decoder side information against the reduced program. Its output is recorded in `verify_decoder_si-2026-10-01.txt`. `verify_fixed_read.py` checks the leakage of a fixed projection direction against a direct conditional-covariance computation, the bounds of Proposition 24 and Theorem 25 on random instances against brute force over directions, and the numbers after Theorem 25. Its output is recorded in `verify_fixed_read-2026-10-01.txt`. `verify_lowdist.py` checks Theorem 15 against the determinant program and a multi-start search on non-commuting instances,

and the rank bound of Proposition 6. Its output is recorded in `verify_lowdist-2026-10-01.txt`. `verify_tilted.py` checks Theorem 7 at the optimum of an independent determinant program on non-commuting instances with partial and full descriptions and holders of every rank, the two-scalar solver of Corollary 9 against that program, the reductions to reverse water-filling and to Theorem 13, and the coupled instance. Its output is recorded in `verify_tilted-2026-10-01.txt`. `verify_scalar_path.py` checks Theorem 18 and the secular equation (11) against the weighted determinant program for scalar described variables with contexts of dimension up to five and holders of every rank (equation (10) had a single root in $[0, 1 - \alpha]$ in every instance), the two limits of the curve, the reduction to Corollary 17, and Corollary 8. Its output is recorded in `verify_scalar_path-2026-10-01.txt`. `verify_mm.py` runs the iteration of Proposition 11 on non-commuting instances with described variables and holders of rank at least two, checks that the cost never increases, and compares the limit with the weighted determinant program. Its output is recorded in `verify_mm-2026-10-01.txt`. The closed-form and scalar algebra is also checked symbolically and formally. `matlab_checks.m` (MATLAB Symbolic Math Toolbox, 11 checks) verifies the root relation and the secular reduction of Theorem 18, the per-direction quadratic of Theorems 13 and 15 and its unique positive root, the identity and monotonicity behind Proposition 24, that the secular equation for $p = 2$ is $-P(g)$ of Corollary 17 over a common denominator, the identity for the tilt after Theorem 7 for a rank-one holder, and the scalar majorizer of Proposition 11. The Lean 4 module `lean/ObservationTheory/RateLeakage.lean` (Mathlib, ten lemmas, no `sorry`, only the standard axioms) machine-checks the same scalar steps and the generic majorization-minimization inequality. The matrix-analytic results, Theorems 5 and 7, Proposition 6, and the convergence part of Proposition 11, are not formalized. Both outputs are recorded in `symbolic_and_lean-2026-10-02.txt`. No proof depends on them. The instances illustrate theorems proved in the text.

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